Separating Common-Factor from Covariate-Specific Spillovers: An Exact SLX Score-Test Decomposition

Marcos Herrera-Gómez*

*CONICET - CIANECO, National University of Rio Cuarto, Argentina.
Email: mherreragomez@conicet.gov.ar.

Abstract

This paper studies the score test for spatially lagged regressors (WX) in SLX models when the data may also exhibit spatial dependence in the lagged dependent variable or the error term. Building on the exact score identity of Koley (2024), $S_\rho = S_\lambda + \hat{\beta}'_{nc} S_\theta$, we show that the unadjusted SLX score test admits an exact, sample-by-sample decomposition $RS_\theta = PS_\theta + OS_\theta$. PS_θ equals numerically, in every sample, the classical robust lag test RS^*_ρ of Anselin et al. (1996); the orthogonal remainder OS_θ, on $k-2$ degrees of freedom, is the only component orthogonal to the common-factor direction of Burridge (1981). It is asymptotically χ^2_{k-2} without normality and, under normality, has an exact finite-sample null law, a rescaled Beta with an equivalent F statistic at every n. OS_θ tests whether the lagged regressors enter only through the aggregated column $WX_{nc}\beta_{nc}$; non-rejection bounds, rather than rules out, spillovers along that direction. At $k = 2$ the remainder is undefined and the battery reduces to the classical tests. Across five cross-sections, the decomposition separates common-factor effects in Ertur–Koch, Columbus, and Georgia from covariate-specific spillovers in St. Louis and Baltimore.

JEL Classification: C12, C15, C21.
Keywords: spatial econometrics, score test, SLX model, robustification, common factor.

Generative AI tools were used extensively throughout the preparation of this manuscript and its accompanying R package, including drafting and revising text, exploring and structuring arguments, aiding in the verification of algebraic derivations, and assisting in the development and testing of the code. The author reviewed all AI-assisted output, independently verified all derivations, proofs, and numerical results, made all substantive scientific decisions, and takes full responsibility for the final manuscript and code. See Declarations for further detail.

1 Introduction

A score (Lagrange multiplier) test can be locally robust to a nuisance parameter, in the sense of Bera and Yoon (1993). However, this robustness can fail when the tested and nuisance directions are exactly linked at the null. In that case, the Bera–Yoon correction does not simply reduce power in one direction. It removes that direction from the test. The remaining directions, if any, can still contain information and can be tested separately.

We study this case in a setting where it arises naturally and has empirical relevance. The setting consists of three nested alternatives to a linear regression that share a common auxiliary matrix. The key feature is more general. An exact score identity can make the joint information matrix singular. This creates two components: one absorbed by the nuisance direction and one orthogonal component that remains informative. This feature can arise in locally misspecified tests whenever a common instrument or auxiliary regressor set links different directions of departure from the null (cf. Bera et al., 2010).

The same construction appears outside spatial econometrics whenever tested and nuisance directions share an auxiliary regressor set. The original locally robust corrections of Bera and Yoon (1993) test serial correlation against ARCH and conversely; Baltagi and Li (1991, 1995) and Bera et al. (2001) extend the same construction to panel data, testing first-order serial correlation against random individual effects and conversely. In each of these cases both the tested and the nuisance parameter are scalar, so the correction can only inflate or deflate power uniformly, and no orthogonal remainder is available to isolate. A vector-valued instance of the same identity should arise in panel tests of higher-order serial correlation, AR(p) with $p \geq 2$, against random individual effects: the equicorrelation that random effects induce loads equally on every lag, so its direction within the space of the p autoregressive coefficients coincides with the equal-weighted direction $(1, \ldots, 1)'$, exactly as β_{nc} does here. Whether this configuration yields the same exact decomposition into an annihilated common-factor component and a lower-dimensional orthogonal remainder is, to our knowledge, an open question.

Readers mainly interested in the general score-test result can go directly to Sections 3-4. The rest of the Introduction and Section 6.2 describe the spatial-econometric setting in which the identity was first noted. They also illustrate the resulting decomposition using real data.

The specific-to-general strategy that underlies much of applied spatial econometrics begins from OLS and uses Moran's I together with the simple Lagrange multiplier, equivalently Rao score, tests of Anselin (1988) to detect spatial dependence. The robust tests of Anselin et al. (1996) then discriminate between a spatial lag model (SLM), a spatial error model (SEM), or both. The broader relationship between this specific-to-general strategy and its general-to-specific alternatives is revisited by Mur and Angulo (2009) and, more recently, by Anselin et al. (2026).

Within this diagnostic tradition, the spatial lag of X (SLX) model occupies a somewhat different position. Gibbons and Overman (2012) and Halleck Vega and Elhorst (2015) argue that the SLX specification,

$$y = X\beta + WX_{nc}\theta + e,$$

where WX_{nc} collects the non-constant regressors, obtained by removing the intercept from X, W is the spatial weight matrix, and e is spatially uncorrelated, can be treated as a natural point of departure rather than an afterthought, and Halleck Vega and Elhorst (2015), in particular, incorporate SLX into model selection through model estimation.

A score-based diagnostic offers a complementary route, since it asks whether the restricted OLS fit already contains evidence of spatially lagged regressors before any enlarged spatial model is estimated. This diagnostic perspective was developed by Koley and Bera (2024) within the spatial Durbin model, $y = \rho Wy + X\beta + WX_{nc}\theta + e$. They derive the joint score test of $\rho = 0$ and $\theta = 0$, the corresponding marginal tests, and the Bera and Yoon (1993) adjustments that make each statistic robust to local departures in the other direction, obtaining a robust lag test and

a robust SLX test together with the additivity relation $RS_{(\rho,\theta)} = RS_\rho + RS^*_{\theta|\rho} = RS^*_{\rho|\theta} + RS_\theta$. The SLX component is thereby brought inside the formal diagnostic framework.

Once the spatial lag, spatial error, and SLX alternatives are evaluated jointly under a common spatial weight matrix, the natural encompassing framework is the General Nesting Spatial (GNS) model,

$$y = \rho W y + X\beta + WX_{nc}\theta + u, \qquad u = \lambda W u + e.$$

Here the GNS serves as an encompassing diagnostic device and not as a model to be estimated. When the same W is used for the spatial lag, the spatial error, and the spatially lagged regressors, the well-known non-identification of the GNS enters the analysis through the structure of the joint score system. A specific question follows: which component of the SLX score is involved in this identification relation, and what complementary information remains available for diagnosing SLX effects.

The answer rests on an exact score identity established by Koley (2024). Let S_ρ, S_λ and S_θ denote the scores for the spatial lag, spatial error, and SLX parameters, each evaluated at the same OLS fit of $y = X\beta + u$ under the joint null $H_0 : \theta = \rho = \lambda = 0$. Writing $\hat{\beta}_{nc}$ for the OLS estimator of the non-constant coefficients, and taking W row-standardized and common to the three spatial components,

$$S_\rho = S_\lambda + \hat{\beta}'_{nc} S_\theta. \tag{1}$$

The identity follows from the OLS first-order conditions alone, so it holds in every sample, for any n and any W satisfying $W\iota = \iota$, without normality and without asymptotics. Two results follow immediately. First, the joint information matrix of the three statistics is singular, with one missing dimension. Second, the Bera and Yoon (1993)-adjusted scores for ρ and λ are zero. However, this does not mean that all the information in the SLX score is lost. The relation involves only the component $\hat{\beta}'_{nc} S_\theta$; the remaining part of S_θ is not affected by the identification problem.

This distinction is our starting point. The SLX coefficient space can be divided into two parts: the direction associated with $\hat{\beta}_{nc}$, which is involved in the score relation of equation (1), and a complementary $(k-2)$-dimensional subspace, which is not. We focus on this latter subspace and define the statistic OS_θ as a standard quadratic form. The direction associated with $\hat{\beta}_{nc}$ corresponds to the restriction $\theta = \tau\beta_{nc}$. Under this restriction, the $k-1$ columns of WX_{nc} enter only through the aggregated column $WX_{nc}\beta_{nc}$. This is the common-factor component of the SLX effect, while OS_θ tests the remaining $k-2$ restrictions. The statistic is asymptotically χ^2_{k-2} without assuming normality. Under normality, its null distribution is exact for every finite n and is a rescaled Beta distribution, with an equivalent ordinary F statistic OF_θ (Section 4.1).

It is important to stress that the null hypothesis of the OS_θ test is the common-factor restriction $\theta \in \text{span}(\beta_{nc})$, not $\theta = 0$. Thus, a rejection indicates that the SLX effects contain covariate-specific spillover effects that cannot be represented by the common-factor direction. A non-rejection, in contrast, is consistent with the SLX effects being fully explained by this direction, even when $\theta = \tau\beta_{nc}$ and τ is arbitrarily large.

The resulting decomposition is exact and additive. In every sample,

$$RS_\theta = PS_\theta + OS_\theta, \qquad RS^*_{\theta|\rho} = RS^*_\lambda + OS_\theta, \tag{2}$$

where RS_θ is the unadjusted score test for the exclusion of WX_{nc} of Koley and Bera (2024), PS_θ is numerically identical to RS^*_ρ, the classical robust lag test of Anselin et al. (1996), RS^*_λ is the corresponding robust error test, and $RS^*_{\theta|\rho}$ is the SLX diagnostic of Koley and Bera (2024) adjusted against ρ alone. These are algebraic identities, not asymptotic equivalences. PS_θ is a one-degree-of-freedom score test of $\hat{\beta}'_{nc}\theta = 0$, directed at the common-factor combination $\beta'_{nc}\theta$ and not at θ as a whole. The remaining orthogonal component of the score is $\tilde{S}_\theta$, and OS_θ is its squared length, taken in the information metric of the score rather than as ordinary Euclidean distance: the only part of the SLX score not captured by the one-dimensional projection. Thus,

OS_θ is the sole increment contributed by the SLX branch beyond the classical battery. The explicit forms are collected in Theorem 1.

The boundary with the prior literature is sharp. The identity (1) is due to Koley (2024) and the unadjusted SLX score RS_θ to Koley and Bera (2024); the singularity of the joint information and the vanishing of the jointly adjusted ρ and λ scores (Proposition 2) are immediate consequences of the identity, restated here only because they underlie the decomposition. What the decomposition adds beyond Koley (2024) is that the SLX score survives the same joint adjustment along its remaining $(k-2)$ directions (our Proposition 3), rather than being annihilated by the singularity. Our contribution is that the identity induces an exact decomposition of the SLX score test into a one-dimensional component identical to the classical robust lag test and a $k-2$-dimensional residual component, together with the residual's null distribution (exact under normality, asymptotic without it), finite-sample implementation, robustness to spatial-error contamination, and relation to the common-factor test.

Three further results follow. When $k = 2$ there is no restriction left beyond the common-factor direction, OS_θ has zero degrees of freedom and is not defined, and the SLX branch adds no degrees of freedom to the classical battery (Corollary 1). When λ is fixed away from zero, asymptotic critical values fail to control size, and Section 5.3 proposes a closed-form rescaling of the variance estimate that restores it under pure error contamination, at a single SARAR fit and with no resampling; the correction is graded rather than universal, and cannot reach the mean-channel distortion that a common W produces when ρ and λ are jointly large. Finally, the SDM of Koley and Bera (2024), which does not include a λWu term by construction, is not subject to (1), so their tests carry genuine information about ρ and θ separately.

The contribution is therefore not the algebraic identity in (1) itself, nor the observation that a projection can be formed from it. Equation (1) identifies one direction of the SLX score that is absorbed by the common-factor relation, but it does not by itself establish what survives the same joint adjustment, nor how that surviving component should be calibrated. Proposition 3 shows that the SLX score survives the joint adjustment precisely along the remaining $k-2$ directions, yielding the orthogonal component OS_θ; the substantive content of the result then lies in characterizing that component. First, Proposition 5 derives its exact finite-sample Beta law and the equivalent F representation, providing a directly implementable finite-sample diagnostic rather than relying only on an asymptotic χ^2_{k-2} approximation. Second, Section 5.2 shows that the behavior of OS_θ under spatial dependence is asymmetric: a local spatial-lag parameter ρ affects the first-order mean in the common-factor direction and is therefore orthogonalized away, whereas spatial-error dependence through λ enters the variance of the score and survives the projection, producing a distinct size distortion. This mean–variance distinction explains why OS_θ is robust to ρ but requires an explicit correction under $\lambda \neq 0$. These two results, the exact finite-sample calibration and the mechanism governing robustness to nuisance spatial dependence, are the main substantive gains beyond the score identity in (2). Both results are stated and proved for the general singular-information setting described above, and neither relies on spatial structure beyond what is needed to construct W: Proposition 5 is a statement about the exact null law of a variable-addition-type F statistic once a rank-deficient direction has been projected out, in the tradition of Godfrey and Veall (2000), and the size mechanism of Section 5.2 is a statement about how a first-order mean shift versus a variance shift are treated differently by an orthogonalized score, which is a property of the Bera-Yoon construction itself rather than of spatial econometrics.

Section 2 states the baseline and the assumptions. Section 3 sets out the identity and the adjusted scores. Section 4 gives the decomposition and its null distributions, exact and asymptotic. Section 5 reports the Monte Carlo evidence: the design (Section 5.1), size (Section 5.2), the closed-form size correction (Section 5.3), and power (Section 5.4). Section 6 illustrates the battery on a synthetic contrast and, for five empirical datasets, we distinguish common-factor effects from covariate-specific spillovers. Section 7 collects what remains open. All proofs are

collected in the appendix.

2 Model, assumptions, and the parameter partition

All tests are constructed under the null model:

$$y = X\beta + u, \qquad u \sim i.i.d.(0, \sigma^2 I), \tag{3}$$

with $X = [\boldsymbol{\iota}, X_{nc}]$ of order $n \times k$. Three departures from (3) are of interest, each a model in its own right, identified without reference to the other two: the spatial lag model (SLM) $y = \rho Wy + X\beta + u$, the spatial error model (SEM) $y = X\beta + u$ with $u = \lambda Wu + e$, and the SLX model $y = X\beta + WX_{nc}\theta + u$. Every statistic below is a score of one of these three, evaluated at the OLS fit of (3). It is convenient, though not strictly required, to specify all three concurrently through the GNS model,

$$y = \rho Wy + X\beta + WX_{nc}\,\theta + u, \qquad u = \lambda Wu + e. \tag{4}$$

The GNS provides the encompassing local framework in which the three score directions can be represented jointly; the results below concern score-based inference at the null and do not require joint estimation of the ρ, λ, and θ, and every hypothesis imposes a restriction on the identified baseline (3). To assist the reader with the notation, Appendix A contains a table listing each element to be used.

The same W governs the spatial lag, spatial error, and spatial lags of the regressors. Because W is row-standardized, $W\boldsymbol{\iota} = \boldsymbol{\iota}$ and the lagged intercept is not separately identified; θ is therefore attached to X_{nc} throughout.

Assumption 1. *W has zero diagonal and is row-standardized. The row and column sums of W and of $(I - \rho W)^{-1}$ and $(I - \lambda W)^{-1}$ are bounded uniformly in absolute value, uniformly in n, for ρ, λ in a compact subset of the parameter space containing zero in its interior (Kelejian and Prucha, 1998, 1999).*

Two conditions do the work: the intercept in X gives $\hat{e}'\iota = 0$, and $W\iota = \iota$ makes the lagged intercept unidentified. Row-standardization is a standard way to secure $W\iota = \iota$, but the identity requires only this property; spectral normalization generally violates it, and islands (zero rows) create a second exception.

Assumption 2. *e_i are i.i.d. with mean zero, variance $\sigma^2 > 0$, and finite fourth moment. Section 3 additionally assumes normality where the information matrix is computed; Section 5.2 reports what happens when this fails.*

The *i.i.d.* assumption excludes heteroskedasticity. The exact finite-sample distribution of Proposition 5 relies on homoskedasticity; the Monte Carlo of Section 7 nonetheless keeps the size of OS_θ close to nominal under covariate-driven heteroskedasticity, up to a 90/10 standard-deviation ratio of 13, and under non-normal (lognormal) errors, so homoskedasticity binds the exact finite-sample law rather than the empirical size.

Assumption 3. *The elements of X are uniformly bounded constants, $n^{-1}X'X$ converges to a finite nonsingular limit, and $n^{-1}X_{nc}'W'MWX_{nc} \to Q_{\theta\theta}$, finite and positive definite, with $M = I - X(X'X)^{-1}X'$.*

Positive definiteness of $Q_{\theta\theta}$ fails when WX_{nc} is nearly in the column space of X, in which case the score test for θ has low power regardless of the adjustment; in practice the condition number of $X_{nc}'W'MWX_{nc}$ should be checked. The joint null $H_0 : \theta = \rho = \lambda = 0$ returns (3), and all scores below are evaluated at its OLS fit, $\hat{\beta} = (X'X)^{-1}X'y$, $\hat{e} = My$, $\hat{\sigma}^2 = \hat{e}'\hat{e}/n$.

3 The identity, and which adjusted scores vanish

Each statistic below is the score of one of the three models of Section 2, evaluated at the OLS fit of (3). Since W has zero diagonal, $\mathrm{tr}(W) = 0$, and the three reduce to

$$S_\theta = \frac{X_{nc}' W' \hat{e}}{\hat{\sigma}^2}, \qquad S_\rho = \frac{\hat{e}' W y}{\hat{\sigma}^2}, \qquad S_\lambda = \frac{\hat{e}' W \hat{e}}{\hat{\sigma}^2}. \tag{5}$$

Differentiating the log-likelihood of (4) in the three parameters at the joint null returns the same three expressions; nothing below depends on which route produced (5).

Proposition 1 (Koley, 2024)**.** *Under H_0, $S_\rho = S_\lambda + \hat{\beta}_{nc}' S_\theta$ exactly, for every sample realization.*

Proof. $y = X\hat{\beta} + \hat{e}$ identically, so $\hat{e}' W y = \hat{e}' W X \hat{\beta} + \hat{e}' W \hat{e}$. Decompose $X\hat{\beta} = \boldsymbol{\iota}\hat{\beta}_1 + X_{nc}\hat{\beta}_{nc}$ and use $W\boldsymbol{\iota} = \boldsymbol{\iota}$ to get $\hat{e}' W X \hat{\beta} = \hat{\beta}_1 \hat{e}'\boldsymbol{\iota} + \hat{\beta}_{nc}' X_{nc}' W' \hat{e}$. The first term vanishes by $\hat{e}'\boldsymbol{\iota} = 0$. Divide by $\hat{\sigma}^2$. □

The result is due to Koley (2024) and is restated here because everything below is an inference from it. It is algebra: it holds for any n, any W with $W\boldsymbol{\iota} = \boldsymbol{\iota}$, and any realization, requiring neither normality nor asymptotics.

Every adjustment below is built from the information of (θ, ρ, λ) partialled on (β, σ^2), written $J_{(\theta,\rho,\lambda).1}$. Under normality, and using $\mathrm{tr}(W) = 0$, its blocks are

$$J_{\theta\theta.1} = \frac{X_{nc}' W' M W X_{nc}}{\sigma^2}, \quad J_{\theta\rho.1} = J_{\theta\theta.1}\beta_{nc}, \quad J_{\theta\lambda.1} = 0, \tag{6}$$

$$J_{\rho\rho.1} = \beta_{nc}' J_{\theta\theta.1} \beta_{nc} + T_W, \quad J_{\lambda\lambda.1} = J_{\rho\lambda.1} = T_W, \quad T_W = \mathrm{tr}(W'W + W^2), \tag{7}$$

with $J_{\theta\theta.1}$ of order $(k-1)\times(k-1)$. The diagonal blocks are the information of each model separately, the last two reproducing Anselin et al. (1996). $J_{\theta\rho.1} = J_{\theta\theta.1}\beta_{nc}$ because partialling on β removes the intercept term from $WX\beta$, exactly as $\hat{e}'\boldsymbol{\iota} = 0$ removes it above; $J_{\rho\lambda.1} = J_{\lambda\lambda.1}$ because $S_\rho - S_\lambda$ is linear in u while S_λ is quadratic, and the two are uncorrelated. It is from these two relations that Proposition 2 reads the singularity.

Proposition 2. *Suppose $J_{\theta\theta.1}$ is nonsingular and $T_W \neq 0$. Then $J_{(\theta,\rho,\lambda).1}$ is exactly singular with rank deficiency 1 and null space spanned by $c = (-\beta_{nc}', 1, -1)'$, in the ordering (θ, ρ, λ).*

Proof. That c lies in the null space is direct verification with (6)-(7): the θ block of $J_{(\theta,\rho,\lambda).1}c$ is $-J_{\theta\theta.1}\beta_{nc} + J_{\theta\rho.1} - J_{\theta\lambda.1} = 0$; the ρ row gives $-\beta_{nc}' J_{\theta\theta.1}\beta_{nc} + (\beta_{nc}' J_{\theta\theta.1}\beta_{nc} + T_W) - T_W = 0$; the λ row gives $0 + T_W - T_W = 0$. For the converse, let $v = (a', b, d)'$ satisfy $J_{(\theta,\rho,\lambda).1}v = 0$. The θ block reads $J_{\theta\theta.1}(a + b\beta_{nc}) = 0$, so $a = -b\beta_{nc}$ by nonsingularity; the λ row reads $T_W(b + d) = 0$, so $d = -b$; the ρ row is then $b(-\beta_{nc}' J_{\theta\theta.1}\beta_{nc} + J_{\rho\rho.1} - T_W) = 0$ for every b, imposing no further restriction. Hence $v = b\,c$, and the null space is one-dimensional. □

Nonsingularity of $J_{\theta\theta.1}$ is the SLX rank condition, MWX_{nc} of full column rank, which Assumption 3 imposes in the limit; that $T_W > 0$ follows from Assumption 1. Proposition 1 makes S_ρ an exact linear combination of S_λ and S_θ in every sample, and $k+1$ statistics tied by one such relation cannot have a full-rank second-moment matrix. What follows is the consequence for the Bera and Yoon (1993) adjustment.

Proposition 3 (Which adjusted scores vanish)**.** *Let $\hat{\beta}_{nc} \neq 0$, which holds with probability one, and let the hypotheses of Proposition 2 hold. Adjust each score against the other two, in the sense of Bera and Yoon (1993), using the partitioned information* (6)-(7) *in plug-in form. Then, in every sample,*

$$\tilde{S}_\rho = 0, \qquad \tilde{S}_\lambda = 0, \qquad \tilde{S}_\theta = S_\theta - \frac{\hat{J}_{\theta\theta.1}\hat{\beta}_{nc}\,\hat{\beta}_{nc}' S_\theta}{\hat{\beta}_{nc}' \hat{J}_{\theta\theta.1}\hat{\beta}_{nc}}. \tag{8}$$

The third is zero for all samples if and only if $k = 2$. For $k \geq 3$ it vanishes only on the event $S_\theta \in \operatorname{span}(\hat{J}_{\theta\theta.1}\hat{\beta}_{nc})$, of probability zero, and it satisfies $\hat{\beta}'_{nc}\tilde{S}_\theta = 0$, so it lies in the ordinary orthogonal complement of $\hat{\beta}_{nc}$ in $\mathbb{R}^{k-1}$, of dimension $k-2$.

Proof. See Appendix B. □

The asymmetry has a simple explanation: identity (1) expresses each of S_ρ and S_λ as an exact linear function of the other two scores, so conditioning removes them completely, while it fixes only the single combination $\hat{\beta}'_{nc}S_\theta = S_\rho - S_\lambda$ of S_θ, leaving the remaining $k-2$ directions unrestricted. This surviving subspace is the ordinary Euclidean complement of $\hat{\beta}_{nc}$, not the $\hat{J}_{\theta\theta.1}$-orthogonal complement $\{v : \hat{\beta}'_{nc}\hat{J}_{\theta\theta.1}v = 0\}$ used by OS_θ below; the two coincide only when $\hat{\beta}_{nc}$ is an eigenvector of $\hat{J}_{\theta\theta.1}$, and substituting one for the other changes the statistic considerably.

The zero block $J_{\theta\lambda.1} = 0$ means S_θ is uncorrelated with S_λ under H_0, but it does not make the joint (ρ, λ) adjustment equivalent to the adjustment for ρ alone, since $J^{-1}_{(\rho\lambda).1}$ is not diagonal; the joint correction involves $S_\rho - S_\lambda$ and differs from the $S^\dagger$ statistic of Koley and Bera (2024) both in the score used and in the denominator, and Theorem 1 shows the difference is exactly RS^*_λ. Consequently, the robustness of OS_θ to a local drift $\lambda_n = c/\sqrt{n}$ comes from $J_{\theta\lambda.1} = 0$, not from ignoring the λ adjustment: S_θ has zero local noncentrality in the λ direction from the start. The over-rejection reported in Section 5.2 when λ is fixed away from zero is a second-order variance effect under a non-local alternative, not evidence against local robustness.

Proposition 4. *Let $\hat{J}^*_{\theta\theta.1} = \hat{J}_{\theta\theta.1} - J_{\theta,(\rho\lambda).1}J^{-1}_{(\rho\lambda).1}J_{(\rho\lambda),\theta.1}$ be the matrix the Bera and Yoon (1993) adjustment of* (8) *produces. Then*

$$\hat{J}^*_{\theta\theta.1} = \hat{J}_{\theta\theta.1} - \frac{\hat{J}_{\theta\theta.1}\hat{\beta}_{nc}\,\hat{\beta}'_{nc}\hat{J}_{\theta\theta.1}}{\hat{\beta}'_{nc}\hat{J}_{\theta\theta.1}\hat{\beta}_{nc}}, \tag{9}$$

which has rank $k-2$ with kernel spanned by $\hat{\beta}_{nc}$. Let $\hat{P}$ be any $(k-1)\times(k-2)$ matrix whose columns span the $\hat{J}_{\theta\theta.1}$-orthogonal complement of $\hat{\beta}_{nc}$, $\hat{P}'\hat{J}_{\theta\theta.1}\hat{\beta}_{nc} = 0$, and define

$$OS_\theta = (\hat{P}'S_\theta)'(\hat{P}'\hat{J}_{\theta\theta.1}\hat{P})^{-1}(\hat{P}'S_\theta) \;=\; (\tilde{S}_\theta)'(\hat{J}^*_{\theta\theta.1})^{+}\tilde{S}_\theta, \tag{10}$$

invariant to the choice of $\hat{P}$. $\hat{P}'\hat{J}_{\theta\theta.1}\hat{P}$ is of full rank $k-2$ and positive definite, so the first expression is an ordinary quadratic form requiring no generalized inverse; the second recovers the same number from the pseudoinverse of $\hat{J}^*_{\theta\theta.1}$ and is stated for continuity with the Bera and Yoon (1993) literature. The statistic vanishes exactly when $S_\theta \in \operatorname{span}(\hat{J}_{\theta\theta.1}\hat{\beta}_{nc})$, of probability zero for $k \geq 3$; the restriction it targets is stated separately, in Remark 1.

Proof. See Appendix B. □

Remark 1 (What the statistic targets)**.** *The restriction tested by OS_θ is $H_0^{OS} : \theta \in \operatorname{span}(\beta_{nc})$, that is, $\theta = \tau\beta_{nc}$ for some scalar τ, under which* (11) *below holds. The local noncentrality of OS_θ is given by*

$$\theta' J_{\theta\theta.1}P(P'J_{\theta\theta.1}P)^{-1}P'J_{\theta\theta.1}\theta,$$

which is nonnegative and vanishes if and only if $\theta \in \operatorname{span}(\beta_{nc})$, under the stated rank conditions. The statistic imposes this restriction through the estimated direction $\hat{\beta}_{nc}$. Under Assumption 3, $\hat{\beta}_{nc} \xrightarrow{p} \beta_{nc}$, so $\hat{P} \xrightarrow{p} P$. Hence, under H_0^{OS}, the local noncentrality induced by estimating the common direction vanishes asymptotically.

This blindness is exact only in the limit. At finite n the estimated direction $\hat{\beta}_{nc}$ does not coincide with its population counterpart, so even under $\theta = \tau\beta_{nc}$ the statistic carries a small nonzero noncentrality, of order $O_p(\tau^2)$ and flat in n (Section 7). Its effect on rejection

is numerically negligible: in the aligned direction of Figure 2, and under $\theta = \tau\beta_{nc}$ in Table 9, OS_θ stays within a few tenths of a percentage point of the nominal level across the range of δ. The larger deviations from nominal size that OS_θ does exhibit are not a property of this common-factor ray but of contamination in the nuisance directions: they arise under $\rho \neq 0$ and, more persistently, under $\lambda \neq 0$, where the $\hat{\sigma}^2$ inflation of Section 5.2 pushes rejection well above 5%. Both channels are absent by construction from the $\rho = \lambda = 0$ design of Figure 2.

The null of OS_θ is a restriction inside the SLX model, and it is worth reading it there. Under $\theta \in \text{span}(\beta_{nc})$, that is $\theta = \tau\beta_{nc}$, the SLX model $y = X\beta + WX_{nc}\theta + u$ becomes

$$y = X\beta + \tau\, WX_{nc}\beta_{nc} + u, \tag{11}$$

in which the $k-1$ spatially lagged regressors enter only through the single aggregated column $WX_{nc}\beta_{nc}$. What OS_θ tests is whether that reduction is tenable, and both hypothesis and alternative live entirely within the SLX family. This is why OS_θ has $k-2$ degrees of freedom and does not exist at $k = 2$. A local departure in ρ moves the mean of y by $\rho WX\beta$ to first order, which is the column $WX_{nc}\beta_{nc}$ of (11) up to the intercept term, so (11) is observationally the same to first order as a small spatial lag: the common-factor restriction $\theta = -\rho\beta$ of Burridge (1981) read from the SLX side. Burridge (1981) works throughout with one regressor, the $k = 2$ case where OS_θ does not exist; the content isolated here lives in the extension he sets aside. The connection is exact and to the score form: the Lagrange multiplier common factor statistic LM_{COM} of Mur and Angulo (2006), evaluated at $\hat{\rho} = 0$ where the restricted SEM fit collapses to OLS, satisfies $LM_{COM}|_{\rho=0} = RS_\lambda + RS_\theta = RS_\lambda + PS_\theta + OS_\theta$ in every sample under $W\boldsymbol{\iota} = \boldsymbol{\iota}$, with $1 + 1 + (k-2) = k$ degrees of freedom. The block common-factor test thus splits into RS_λ, $PS_\theta = RS^*_\rho$ and OS_θ: two summands already in routine use and one, OS_θ, attributable neither to error dependence nor to a lag. The identity was verified to 10^{-14} for $k \in \{3, 4, 5\}$ under symmetric and row-standardized asymmetric W.[1]

Two further consequences follow. First, $k = 2$ is far from pathological: it covers every application with one substantive regressor, and Section 6.2 shows it covers Ertur and Koch's own preferred specification. Second, the kernel depends on β_{nc}, which is estimated: when $\beta_{nc} = 0$ in the population, the degrees of freedom of the *population* object jump discontinuously from $k-2$ to $k-1$, but the computed statistic does not inherit the jump, since $\hat{\beta}_{nc}$ is nonzero with probability one and $\hat{J}^*_{\theta\theta.1}$ has rank exactly $k-2$ in every sample. Size is likewise unaffected near $\beta_{nc} = 0$, since $\hat{\beta}_{nc}$ is independent of $\hat{e}$ under normality; what degrades is the target rather than the calibration.

Remark 2 (Partial and sparse WX)**.** *When only a subset of the regressors carries a spatial lag, so that θ is supported on a proper subset of $\{1, \ldots, k-1\}$, alignment is a property of the whole vector θ in the $J_{\theta\theta.1}$ metric and not of the individual regressors. A single lagged regressor $\theta = \tau e_j$ lies in* span(β_{nc}) *only if $\beta_{nc} \propto e_j$, which is degenerate, and is $J_{\theta\theta.1}$-orthogonal to β_{nc} only if $\beta'_{nc}J_{\theta\theta.1}e_j = 0$; away from these two edges it carries both an aligned and an orthogonal component and moves PS_θ and OS_θ at once. A sparse θ is therefore generically off the common-factor line, since a vector supported on a strict subset cannot be proportional to a full-support β_{nc}, so OS_θ has power against exactly the covariate-specific spillovers that a partial WX represents, and the common-factor restriction $\theta = -\rho\beta_{nc}$ is violated by construction unless β_{nc} shares the same support. Two qualifications bound this reading. OS_θ is an omnibus test over the $k-2$ orthogonal directions: it detects that off-line content is present but does not localise which regressor carries it, which requires the coordinate-wise SLX estimates rather than the score; and against a sparse alternative concentrated in a single direction it spreads $k-2$ degrees of freedom over a low-dimensional departure, so it remains consistent but is dominated in power by a test directed at that Wx_j. The decomposition itself is unaffected: the identity $S_\rho = S_\lambda + \hat{\beta}'_{nc}S_\theta$*

[1] Two blocks of the information matrix reported by Mur and Angulo (2006) are stated for symmetric W; the identity above uses their row-standardized counterparts.

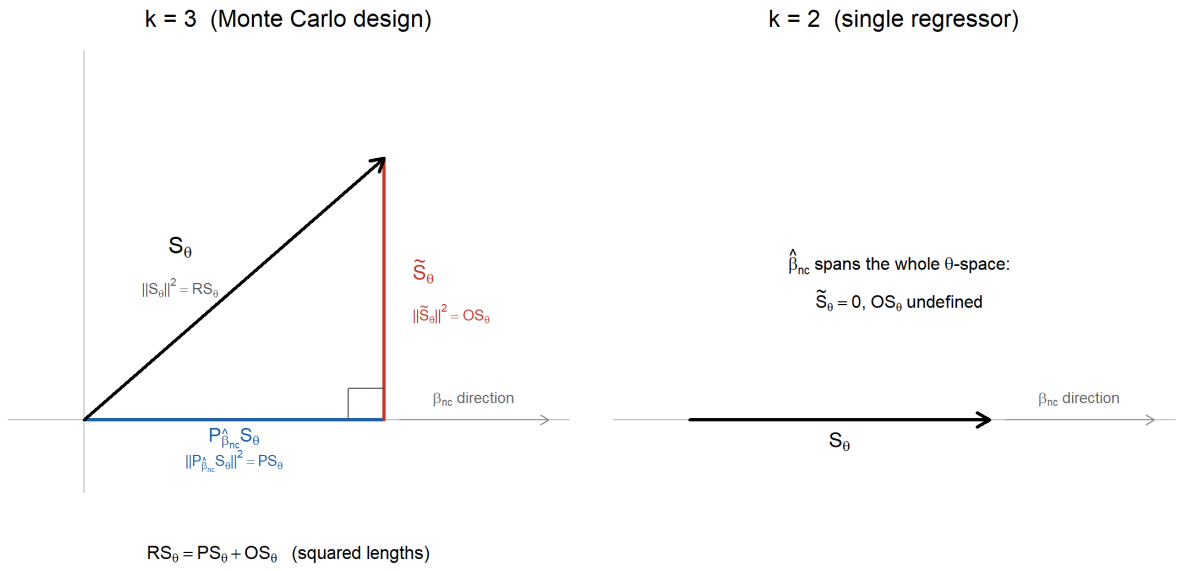


Figure 1: The decomposition $RS_\theta = PS_\theta + OS_\theta$ as a Pythagorean relation in the $\hat{J}_{\theta\theta.1}$ metric, coordinates whitened so the right angle drawn is literal. The three sides are score vectors, and the labelled statistics are their squared lengths: the hypotenuse is the SLX score S_θ ($\|S_\theta\|^2 = RS_\theta$), the horizontal leg its projection $P_{\hat{\beta}_{nc}} S_\theta$ onto the common-factor direction $\hat{\beta}_{nc}$ ($\|\cdot\|^2 = PS_\theta$), and the vertical leg the orthogonal residual $\tilde{S}_\theta$ ($\|\tilde{S}_\theta\|^2 = OS_\theta$). Left: $k = 3$, one representative realization. Right: $k = 2$, where $\hat{\beta}_{nc}$ spans the entire θ-space, so $\tilde{S}_\theta = 0$ and OS_θ is not defined.

is an OLS first-order condition independent of θ and its support, so $RS_\theta = PS_\theta + OS_\theta$ holds unchanged and only the location of the power shifts.

4 The decomposition and the null distribution

Theorem 1 (Exact decomposition)**.** *Under $H_0 : \theta = \rho = \lambda = 0$ and $\hat{\beta}_{nc} \neq 0$, which holds with probability one, in every sample and for every W with $W\iota = \iota$,*

$$PS_\theta = \frac{(\hat{\beta}'_{nc} S_\theta)^2}{\hat{\beta}'_{nc} \hat{J}_{\theta\theta.1} \hat{\beta}_{nc}}, \tag{12}$$

$$RS_\theta = PS_\theta + OS_\theta, \tag{13}$$

$$RS^*_{\theta|\rho} = RS^*_\lambda + OS_\theta, \tag{14}$$

where RS_θ, the unadjusted SLX score test of Koley and Bera (2024), decomposes as in (13) *into PS_θ, numerically identical in every sample to RS^*_ρ, the robust lag test of Anselin et al. (1996), and OS_θ; and RS^*_λ, their robust error test, and $RS^*_{\theta|\rho}$, the SLX test of Koley and Bera (2024) adjusted against ρ alone, decompose likewise as in* (14)*. Each statistic is computable from the OLS fit of* (3) *alone.*

Proof. See Appendix B. □

Equation (13) is Pythagoras in the $\hat{J}_{\theta\theta.1}$ metric: any direction in $\mathbb{R}^{k-1}$ splits RS_θ into a squared projection and a squared orthogonal residual. What Proposition 1 supplies is not the split but the direction: absent it, PS_θ is the squared length of the component of S_θ along $\hat{\beta}_{nc}$, with no reason to prefer that line to any other; with it, $\hat{\beta}'_{nc} S_\theta = S_\rho - S_\lambda$, and PS_θ is the classical robust lag test in every sample. The second identity is $OS_\theta = RS^*_{\theta|\rho} - RS^*_\lambda$: what is left of Koley and Bera (2024)'s spatial Durbin diagnostic once the classical robust error test is taken out.

Identities (13)-(14) were verified numerically to within 10^{-14} for $k \in \{2, 3, 4, 6\}$ and random W, and to eleven decimals on the data of Section 6.2.

Equation (12) carries two readings that must be kept at their own levels. In the sample, PS_θ is an exact, finite-sample one-degree-of-freedom score test of $\hat{\beta}'_{nc}\theta = 0$, numerically identical to RS^*_ρ; the population combination the contrast targets is $\beta'_{nc}\theta$, asymptotically, under consistency of $\hat{\beta}_{nc}$. The practice of consulting RS^*_ρ and RS^*_λ under a common W and using them to identify a "lag" specification follows the standard logic of the LM battery. A common W is required for these statistics to be comparable and for the battery to serve as a model-selection device (Anselin et al., 1996; Mur and Angulo, 2009). The identity shows, however, that under a common W, the battery cannot distinguish a spatially lagged dependent variable from the common-factor combination $\beta'_{nc}\theta$: both are represented by the same score direction. The incremental content of the SLX branch over the classical battery is therefore exactly OS_θ, with $k-2$ degrees of freedom.

A qualification belongs here, since (14) can be misread as separating a neutral λ term from a genuinely new θ term, and only one of the two has that property. Although $J_{\theta\lambda.1} = 0$ makes S_θ and S_λ exactly uncorrelated under H_0, RS^*_λ is not itself locally blind to θ along the common-factor direction: its local noncentrality under $\theta = t\beta_{nc}$ pure equals $t^2\, T_W\, \beta'_{nc} J_{\theta\theta.1} \beta_{nc}/D$, with $D = J_{\rho\rho.1} = \beta'_{nc} J_{\theta\theta.1} \beta_{nc} + T_W$, strictly positive for $t \neq 0$. The channel is S_ρ, not S_λ directly: the Bera and Yoon (1993) adjustment defining RS^*_λ is built from S_ρ, and S_ρ moves with θ by Proposition 1 even though S_λ's own mean does not. OS_θ, by construction, has exactly zero noncentrality in this same direction (Remark 1). The two terms of (14) are therefore not symmetric in what they are sensitive to, and the identity should be read as an exact arithmetic decomposition rather than as isolating, cleanly, how much of $RS^*_{\theta|\rho}$ is attributable to λ.

Corollary 1 (The single-regressor case)**.** *If $k = 2$ and $\hat{\beta}_{nc} \neq 0$, then $RS_\theta = PS_\theta$ and $RS^*_{\theta|\rho} = RS^*_\lambda$ exactly, in every sample, and OS_θ does not exist.*

Corollary 1 follows from Theorem 1 with $OS_\theta = 0$. With one substantive regressor the SLX score battery of Koley and Bera (2024), computed from an OLS baseline in the presence of a possible spatially correlated error, returns the classical Anselin et al. (1996) numbers relabelled.

4.1 Exact finite-sample null distribution

The null distribution of OS_θ is available exactly, at every finite n; it requires normality of e. This is exactness under the complete null $\theta = \rho = \lambda = 0$, not robustness to spatial-error contamination: when $\lambda \neq 0$, OF_θ inherits the size distortion of Section 5.2, and the closed-form rescaling of Section 5.3, not the exact law, is the calibration to report. The asymptotic χ^2_{k-2} approximation of Theorem 2 below is what remains when normality is not assumed.

Proposition 5 (Exact finite-sample null distribution)**.** *Under Assumption 2 with normal errors, at the exact null $\theta = \rho = \lambda = 0$, $\hat{\beta} = (X'X)^{-1}X'y$ is independent of $\hat{e} = Me$. Consequently, writing $q = k - 2$ and $B = OS_\theta/n$ for the statistic of* (10) *rescaled, for every population value of β_{nc} (fixed, drifting as $n^{-1/2}b$, or exactly zero) the finite-sample null law of B is exactly*

$$B \sim \text{Beta}\Big(\tfrac{q}{2},\, \tfrac{n-k-q}{2}\Big), \qquad OF_\theta = \frac{n-k-q}{q} \cdot \frac{B}{1-B} \sim F_{q,\,n-k-q}.$$

Equivalently $B = (SSR_r - SSR_u)/SSR_r$, the residual-sum-of-squares ratio between the OLS regression of y on X alone and on X together with $Z = WX_{nc}\hat{P}$. Both laws hold exactly at every finite n; OS_θ itself is not χ^2_{k-2} at finite n, though $n \cdot B \xrightarrow{d} \chi^2_q$ as $n \to \infty$, recovering Theorem 2 as a special case.

Proof. See Appendix B. □

This is not a repair of the asymptotic argument: it needs no limit, because the Beta law of B is exact at every fixed n regardless of where β_{nc} sits, including arbitrarily close to zero or exactly at it; only the χ^2_{k-2} calibration used by OS_θ is asymptotic. The reason is visible in the numerical rank of $\hat{J}^*_{\theta\theta.1}$: built from the almost-surely-nonzero $\hat{\beta}_{nc}$, that rank is $k-2$ in every replication run, so rejection rates track the Beta-implied rates within Monte Carlo error at every sample size. The argument is Gaussian-specific: repeating the same grid under lognormal errors gives rejection rates within a point of the Gaussian ones in this design, evidence rather than proof, since Proposition 5 is stated under normality only.

The same statistic has a classical, non-pseudoinverse route to the same exact law: regressing y on X and on $Z = WX_{nc}\hat{P}$ and computing the ordinary F statistic for the exclusion of Z, on $k-2$ and $n-k-(k-2)$ degrees of freedom, gives the same contrast without going through this paper's identity or a pseudoinverse. We write it OF_θ; both are deterministic transforms of the Beta variable B, and Section 5.2 verifies this numerically. The equivalence is exact only at the complete null: under $\lambda \neq 0$ the residual is no longer isotropic and nothing of the exactness survives, and Section 5.2 reports what is left. Table 2 shows OF_θ sitting a few tenths of a percentage point below OS_θ at $n = 100$, a gap that narrows with n, since χ^2_{k-2} is the $n \to \infty$ limit of $F_{k-2,n-k-(k-2)}$; both are simultaneously undefined at $k = 2$. A practitioner who wants the exact finite-sample size documented in Section 5.2 should compute OF_θ instead of OS_θ; exactness at the null is not robustness, since OF_θ is no better protected against λ contamination, being a deterministic transform of the same B.

An F test directed at spatially lagged regressors is not itself new. Florax and Folmer (1992) propose one, their FOV statistic, as the F-based branch of a variable-expansion search that adds omitted spatial lags to an OLS specification, as against the Lagrange multiplier branch built on the tests of Anselin (1988). Their statistic is the ordinary variable-addition F on the full WX_{nc} block, so it retains the common-factor direction β_{nc} that OF_θ discards, precisely the direction along which, by (13), a spatial lag enters the SLX score. Their own Monte Carlo evidence is not encouraging. Even in the cross-regressive design, where a spatially lagged regressor is the only departure from OLS, the F-based search is outperformed by its Lagrange multiplier counterpart. Moreover, the latter recovers the true model with a probability that they regard as insufficient when the lagged coefficient is positive. The gain from OF_θ therefore arises precisely where their F test is weakest: by removing the component associated with β_{nc}, OF_θ avoids the confounding that occurs when a conventional F test on WX_{nc} cannot distinguish a genuine SLX effect from a spatial lag.

4.2 The asymptotic null distribution, without normality

Theorem 2. *Under Assumptions 1–3, $H_0 : \theta = \rho = \lambda = 0$, and $\beta_{nc} \neq 0$ fixed with $k \geq 3$, $OS_\theta \xrightarrow{d} \chi^2_{k-2}$, the degrees of freedom being the rank of $J^*_{\theta\theta.1}$ rather than its dimension.*

The proof is in Appendix C. Two steps deserve to be flagged. First, the limiting distribution of $n^{-1/2}S_\theta$ follows from the central limit theorem for linear-quadratic forms of Kelejian and Prucha (2001), whose conditions Assumption 1 is written to supply. Second, OS_θ as defined in (10) is an ordinary quadratic form in a $(k-2)$-vector with a positive definite weighting matrix, so no generalized inverse enters its limit theory. Theorem 2 is stated under Assumption 2, which does not require normality; the following lemma establishes that the covariance blocks it uses are the Gaussian ones whatever the error distribution.

Lemma 1. *Let e_i be i.i.d. with mean zero, variance σ^2, third moment μ_3, fourth moment μ_4, all finite, and let W have a zero diagonal. Then the asymptotic variances and covariances of $(S_\theta, S_\rho, S_\lambda)$ under $H_0 : \theta = \rho = \lambda = 0$ do not depend on μ_3 or μ_4, and coincide with the Gaussian expressions (6)–(7). Consequently Theorem 2 holds under Assumption 2 without normality.*

Proof. See Appendix B. □

The requirement that β_{nc} be fixed and nonzero is not a technicality for the asymptotic route of Theorem 2: along a sequence $\beta_{nc,n} = n^{-1/2}b$ the direction β_{nc} collapses, so the $J_{\theta\theta.1}$-orthogonal complement that defines P has no limit, and Theorem 2 does not, by itself, cover this case. Proposition 5 above does, holding at every n without appeal to any limit and for every population value of β_{nc} including exactly zero, which is one reason for stating the exact law first.

5 Monte Carlo evidence

5.1 Design

Data are generated from (4) with the parameters set by us, and the singularity of Proposition 2 constrains which configurations may be used. Along the null direction $c = (-\beta'_{nc}, 1, -1)'$ distinct configurations generate the same first-order distribution: $\rho = t$ pure and the pair $(\theta = t\beta_{nc}, \lambda = t)$ differ by exactly $t \cdot c$ and are locally indistinguishable, so a configuration with ρ, λ and θ all nonzero does not define a single alternative against which power can be reported. Size and power fall on opposite sides of this constraint. Under the null tested throughout, $\theta = 0$, and c requires all three coordinates to move together, so a size cell with $\theta = 0$ and (ρ, λ) both active lies outside the kernel of J: it is a well-defined bivariate SARAR design, and every size cell below is of this kind. The power study needs $\theta \neq 0$ and is therefore run bivariate: θ against a clean baseline, θ under a single active lag ($\rho = 0.3$), and θ under a single active error ($\lambda = 0.3$), each isolating one contamination channel. Section 7 records the all-three-at-once configuration as a limitation a common W cannot represent as a power alternative.

The design crosses $n \in \{49, 100, 225\}$ on a hexagonal lattice (Anselin and Florax, 1995; Florax and Folmer, 1992), normal and lognormal (centred, variance matched) errors, and ρ, λ grids including negative values, with 5000 replications per cell. Throughout $k = 3$, so OS_θ has one degree of freedom; the $k = 2$ case is degenerate by Corollary 1 and needs no simulation. The regressor matrix $X = [\iota, X_{nc}]$ is redrawn inside the replication loop, with $k - 1$ columns of X_{nc} *i.i.d.* $\mathcal{N}(0, 1)$, independent of W; $\beta = (1, 1, 2)'$, so $\beta_{nc} = (1, 2)'$; $\sigma^2 = 1$; W the row-standardized hexagonal matrix. Drawing X_{nc} independently of W gives $E[X'WX_{nc}] = 0$:[2] the restricted OLS fit that estimates the common-factor direction is then consistent for its population target even under a fixed, non-local alternative. This property is used in the reading of the size mechanism below and is flagged again in Section 6.2, where the data do not have it.

The hexagonal lattice gives every interior cell exactly six neighbours at equal centre-to-centre distance, following the spatial simulation literature (Anselin and Florax, 1995; Florax and Folmer, 1992). The structure of W enters the mechanism through T_W and $X'_{nc}W'MWX_{nc}$ only; every identity above is algebraic in these and holds for any W with $W\iota = \iota$. What can depend on W is the magnitude, not the existence, of the size distortions of Section 5.2; both a k-nearest-neighbours check and the real-geography weight matrices of Section 6.2 give the same qualitative reading.

Since OS_θ has no local power against $\theta \propto \beta_{nc}$ at any magnitude, and at finite n only an $O_p(\tau^2)$ residual, a power study varying only the magnitude of $\theta = \text{frac} \cdot \beta_{nc}$ would report nominal rejection almost everywhere by construction. We instead set $\theta = \delta v$ with v a unit vector either aligned with β_{nc} or $J_{\theta\theta.1}$-orthogonal to it, varying signed δ over a fine grid.

A standard F test of the exclusion of WX_{nc} appears throughout as a benchmark, written F_θ: the variable-addition test of Godfrey and Veall (2000), on $(k-1,\ n-2k+1)$ degrees of freedom. It is the benchmark for RS_θ, not the OF_θ of Section 4.1: it retains the common-factor direction, spends one more restriction, and exists at $k = 2$. Neither is immune to the contamination the

[2] Appendix E relaxes this and finds that raising the correlation between the regressors and their spatial lags to 0.85 leaves the size and power of OS_θ essentially unchanged, with only a second-order effect on the estimated common-factor direction.

OLS baseline omits, and the two respond to it differently in finite samples: RS_θ inherits the lag noncentrality through the closed-form channel of (13), while the F statistic responds through the residual sums of squares directly, and the two need not agree away from the null even though Godfrey and Veall (2000) pin them together asymptotically.

5.2 Size

Table 1 reports seven selected cells: the exact null; the pure-lag cells $\rho \in \{-0.3, 0.3, 0.7\}$ at $\lambda = 0$; and the mixed cells $(\rho, \lambda) \in \{(-0.3, 0.7), (0, 0.7), (0.7, 0.7)\}$, spanning both signs of ρ and the hardest error cell $\lambda = 0.7$. The full 7×7 grid, and an extended grid over $k \in \{3, 4, 6\}$, are available in the Supplementary Material.

Table 1: Rejection rate at nominal 5%, $n = 100$, 5000 reps/cell, normal errors, five selected cells. Bold marks values outside the 95% band [0.0440, 0.0560]. Full 7×7 grid at every n in the online appendix.

ρ	λ	RS_θ	OS_θ
0	0	**0.057**	0.053
0.7	0	**1.000**	**0.042**
0	0.7	**0.263**	**0.192**
0.3	0.3	**0.748**	**0.100**
0.7	0.7	**0.907**	**0.153**

RS_θ, unadjusted, over-rejects severely whenever $|\rho| \geq 0.3$, reaching 60-100% at a nominal 5% level regardless of λ: by (13), RS_θ contains PS_θ as a summand, and PS_θ is designed to reject when $\rho \neq 0$. OS_θ corrects this across the tested range of ρ, but not symmetrically: at $\lambda = 0$, rejection is 2.6% at $\rho = -0.3$ against 6.1% at $\rho = 0.3$, a factor-of-two gap holding at every λ tested. Two effects run in opposite directions, one symmetric (the term $\hat{P}$ purges from the numerator inflating $\hat{\sigma}^2$) and one asymmetric (higher-order terms leaking into the numerator, accumulating under $\rho > 0$ and partly cancelling under $\rho < 0$ since W is row-standardized and non-negative). The asymmetry is real, a genuine weakness of OS_θ's size control. Under λ fixed away from zero rejection climbs to 15.1–19.2% at $\lambda = 0.7$: not a local-robustness failure ($J_{\theta\lambda.1} = 0$ makes the adjustment against locally drifting λ exact), but the second-order variance effect of a fixed, non-local λ, traced below to the estimation of σ^2.

Table 2 shows that exactness and robustness answer different questions. Because OF_θ and OS_θ are strictly monotone transforms of the same statistic B, they have the same qualitative response to spatial-error contamination. Under the complete null, both statistics are correctly calibrated at approximately the nominal 5% level. When $\rho = 0.3$ and $\lambda = 0$, however, the unadjusted statistics RS_θ and F_θ reject at 78.9% and 78.1% for $n = 100$, rising to 98.4% for $n = 225$, whereas the orthogonalized statistics remain close to nominal size, with rejection rates of 6.1% and 5.6% at $n = 100$ and 6.3% and 6.2% at $n = 225$. The same contrast is visible at $\rho = 0.7$: the unadjusted statistics reject in all simulations, while OS_θ and OF_θ remain close to the nominal level, with rejection rates between 3.7% and 4.2%.

The table also shows that orthogonalization does not by itself eliminate distortion generated by spatial-error dependence. When $\lambda = 0.7$ and $\rho = 0$, OS_θ and OF_θ reject at 19.2% and 18.5%, respectively, for $n = 100$, and 19.9% and 19.6% at $n = 225$. Thus, the small calibration gap between the two strictly monotone representations is essentially negligible, while the distortion itself persists as the sample size increases. The milder case $\lambda = 0.3$ produces substantially smaller distortions: OS_θ rejects at 8.7% for $n = 100$ and 7.7% for $n = 225$, with corresponding OF_θ rejection rates of 8.2% and 7.4%.

Taken together, these results separate two issues. Orthogonalization removes the severe size distortion induced by spatial-lag contamination in the $\rho > 0, \lambda = 0$ cells, but it does not remove

Table 2: Rejection rate at nominal 5%, normal errors, $k = 3$, five selected cells, $n \in \{100, 225\}$, 5000 replications per cell. RS_θ and OS_θ are read against χ^2_{k-1} and χ^2_{k-2}; F_θ and OF_θ against their exact F distributions. F_θ excludes all of WX_{nc} and is the finite-sample twin of RS_θ; OF_θ excludes $WX_{nc}P$ only and is the exact counterpart of OS_θ (Section 4.1). All four are computed on the same replications, so within each pair the two columns differ through the reference distribution only, not through the sample. The full grid at every $n \in \{49, 100, 225\}$ is in the Supplementary Material.

		unadjusted		orthogonalized	
ρ	λ	RS_θ	F_θ	OS_θ	OF_θ
$n = 100$					
0.0	0.0	**0.057**	0.053	0.053	0.050
0.3	0.0	**0.789**	**0.781**	**0.061**	**0.056**
0.7	0.0	**1.000**	**1.000**	**0.042**	**0.039**
0.0	0.3	**0.111**	**0.106**	**0.087**	**0.082**
0.0	0.7	**0.263**	**0.257**	**0.192**	**0.185**
$n = 225$					
0.0	0.0	0.055	0.053	**0.058**	**0.057**
0.3	0.0	**0.984**	**0.984**	**0.063**	**0.062**
0.7	0.0	**1.000**	**1.000**	**0.038**	**0.037**
0.0	0.3	**0.098**	**0.096**	**0.077**	**0.074**
0.0	0.7	**0.264**	**0.261**	**0.199**	**0.196**

the distortion induced by spatial-error dependence. Under the maintained Gaussian complete-null calibration, OF_θ is therefore the natural exact representation of the orthogonal test; when λ is non-negligible, additional calibration is required.

Why OS_θ controls size against ρ but not against λ

Under lag contamination $y = (I - \rho W)^{-1}(X\beta + e)$, to first order in ρ,

$$E[S_\theta] \;\approx\; \rho\, J_{\theta\theta.1}\, \beta_{nc}. \tag{15}$$

The bias lies in the direction $J_{\theta\theta.1}\beta_{nc}$, exactly what $\hat{P}'$ removes, so the contaminated component of S_θ does not reach $\hat{P}'S_\theta$ except through the estimation error in $\hat{\beta}_{nc}$; under this design, $E[X'WX_{nc}] = 0$ makes that residual channel estimation error rather than bias. Error contamination produces no systematic drift in $E[S_\theta]$ at all, so λ acts through the variance instead: writing $\Omega_\lambda = (I - \lambda W)^{-1}(I - \lambda W')^{-1}$, the exact variance-inflation ratio (if σ^2 were known) gives 23–29% rejection at $n = 100$, overshooting the observed over 18% because $OS_\theta \propto 1/\hat{\sigma}^2$ and $\hat{\sigma}^2$ is itself inflated, $E[\hat{\sigma}^2]/\sigma^2 = \mathrm{tr}(M\Omega_\lambda)/n = 1.584$ at $n = 100$; correcting for this brings the prediction to within one Monte Carlo standard error of the tabled values. The flatness in n is therefore two offsetting effects: the σ^2-known mechanism grows with n and the $\hat{\sigma}^2$ inflation grows at close to the same rate.

Under ρ contamination the picture differs, and in the direction that motivates the orthogonalization. In the baseline design, where $E[X'WX_{nc}] = 0$, the first-order drift of the SLX score lies along $J_{\theta\theta.1}\beta_{nc}$: $E[S_\theta] \approx \rho J_{\theta\theta.1}\beta_{nc}$. The projector therefore removes the component generated by lag contamination to first order. This is visible directly in Table 1: at $\rho = 0.7$ and $\lambda = 0$, the unadjusted statistic rejects in every replication, whereas OS_θ rejects only 4.2% of the time. Thus the dominant size distortion induced by the spatial-lag component is removed by the orthogonalization, although the higher-order terms generated by the spatial-lag filter need not lie in the first-order direction and can leave a residual distortion. The latter is particularly

visible for positive ρ and is one reason why the size control of OS_θ is not exact under non-local lag contamination.

5.3 A closed-form correction for the λ distortion

The $\hat{\sigma}^2$-inflation account above points to a correction. Conditional on X, $a = \hat{P}'(WX_{nc})'\hat{e}$ is exactly Gaussian with variance $\sigma^2\,\hat{P}'(WX_{nc})'M\Omega_\lambda M(WX_{nc})\hat{P}$, so the sandwich statistic that replaces the denominator's implicit $\Omega_\lambda = I$ by $\hat{\Omega}_\lambda = (I - \hat{\lambda}W)^{-1}(I - \hat{\lambda}W')^{-1}$ has the correct central matrix; with σ^2 known it is χ^2_{k-2} in every sample. What fails in the naive plug-in is the scale: retaining $\hat{e}'\hat{e}/n$ estimates $\sigma^2\,\mathrm{tr}(M\Omega_\lambda)/n$, not σ^2, and divides the statistic a second time by the same 1.584 inflation, pushing rejection under $\lambda = 0.7$ to 1–2% rather than to nominal, an oracle in λ included. The repair rescales the variance estimate,

$$\tilde{\sigma}^2 = \hat{e}'\hat{e}/\,\mathrm{tr}(M\hat{\Omega}_\lambda), \tag{16}$$

consistent estimator of σ^2 and equal to $\hat{e}'\hat{e}/(n-k)$ when $\hat{\Omega}_\lambda = I$, with the trace $\mathrm{tr}(M\Omega) = \mathrm{tr}(\Omega) - \mathrm{tr}((X'\Omega X)(X'X)^{-1})$ costing nothing further. Taking $\hat{\lambda}$ from the SARAR fit under $H_0 : \theta = 0$, the rescaled statistic holds nominal size across pure error contamination to 5.1% at $\lambda = 0.7$ against a raw 18.6% ($n = 100$), and under lognormal errors alike; estimating λ is free, matching the oracle to two decimals. The correction is graded rather than universal: under ρ alone $\hat{\lambda}$ is near zero and the rescaling is close to inert, but at $(\rho, \lambda) = (0.7, 0.7)$, both near the stationarity boundary with a common W, the SARAR fit is weakly identified, collapses both estimates toward an interior point ($\hat{\rho} = 0.69, \hat{\lambda} = 0.64$), and the statistic under-rejects at 2.2–2.7%: a variance correction cannot reach what a common lag filter injects through the mean.

Table 3: Size of the rescaled sandwich (16). Rejection rate (%) at nominal 5%, $\theta = 0$, hexagonal W, 5000 replications per cell. "raw" is OS_θ with the uncorrected denominator $\hat{e}'\hat{e}/n$; "resc." is the rescaled statistic with $\hat{\lambda}$ from the SARAR fit under $H_0 : \theta = 0$. Monte Carlo standard error at 5% is 0.31 points. A ten-cell view of the full $7 \times 7 \times n$ grid.

	normal		lognormal	
(ρ, λ)	raw	resc.	raw	resc.
$n = 100$				
0, 0	**5.9**	**5.8**	**5.7**	5.4
0, 0.3	**8.8**	5.5	**8.4**	5.1
0, 0.5	**12.6**	5.2	**12.3**	5.4
0, 0.7	**18.6**	5.1	**18.1**	5.0
0.3, 0	**6.3**	**6.0**	5.4	5.0
0.7, 0	4.7	**4.0**	**3.3**	**3.1**
−0.3, 0	**3.0**	**3.0**	**2.9**	**2.9**
0.3, 0.3	**9.4**	**5.8**	**9.0**	5.1
0.3, 0.7	**18.3**	5.0	**18.1**	4.7
0.7, 0.7†	**15.0**	**2.7**	**14.2**	**2.2**

5.4 Power

The three scenarios below are bivariate by construction, for the reason given in Section 5.1: θ is always the alternative of interest, and at most one of ρ, λ is active alongside it. Under a local alternative $\theta = \theta_n \to 0$, $E[S_\theta] \to J_{\theta\theta.1}\theta$, so the noncentrality of RS_θ is $\theta' J_{\theta\theta.1}\theta$, the textbook quadratic form. For OS_θ the computation is conditional on $\hat{P}$: using $E[\tilde{S}_\theta \mid \hat{P}] = \hat{J}^*_{\theta\theta.1}\theta$, the noncentrality is $\theta'\hat{J}_{\theta\theta.1}\theta - (\hat{\beta}'_{nc}\hat{J}_{\theta\theta.1}\theta)^2/(\hat{\beta}'_{nc}\hat{J}_{\theta\theta.1}\hat{\beta}_{nc})$, exactly RS_θ's noncentrality minus the squared common-factor component. At $\theta = \tau\beta_{nc}$ this is zero in population and

$O_p(\tau^2)$ conditionally, recovering Proposition 4 from the noncentrality formula; away from that direction the two coincide only when θ is $J_{\theta\theta.1}$-orthogonal to β_{nc}.

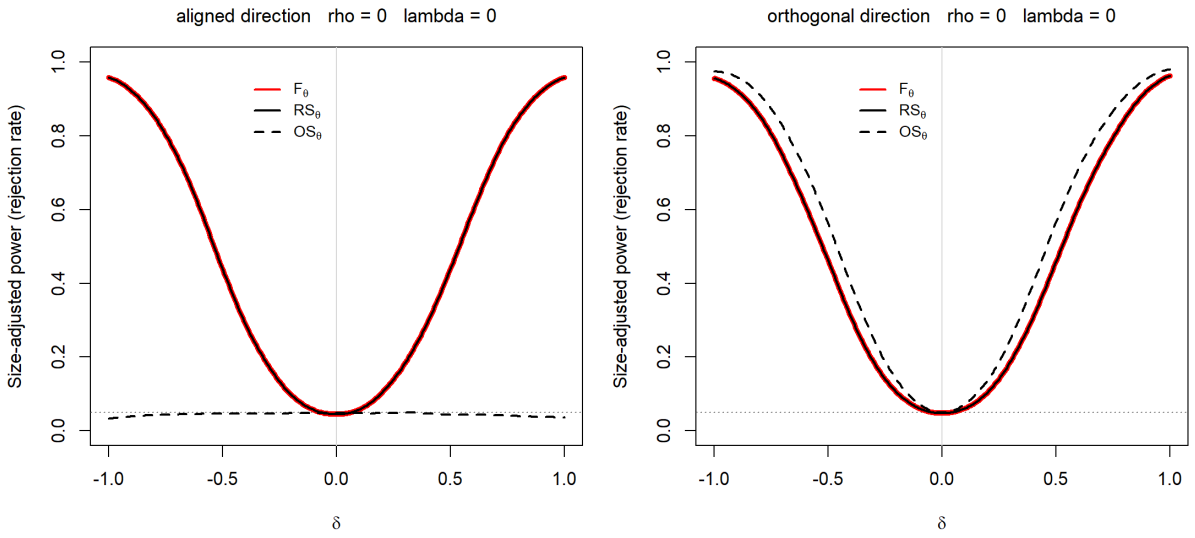


Figure 2: Size-adjusted power, $\rho = 0$, $\lambda = 0$ (no contamination). Left: θ aligned with β_{nc}. Right: orthogonal.

Size-adjusted comparisons use empirical critical values from the matching null cell, a laboratory device not available to a practitioner, who does not know the true (ρ, λ); Section 5.3 supplies the feasible counterpart. In the aligned direction (left panel of Figure 2) OS_θ is flat at 2–7% across the entire range of δ, while F_θ and RS_θ trace the usual symmetric power curve rising above 95%: Proposition 4 made visible. Under $\rho = 0.3$ contamination the pattern is unchanged in the aligned direction, and in the orthogonal direction the advantage of OS_θ widens sharply, reaching 0.71 against 0.22 for F_θ at $\delta = 0.6$. Under $\lambda = 0.3$ the orthogonal gap narrows again (0.59 against 0.49 at $\delta = 0.6$), tracking the size distortions of Section 5.2. In the orthogonal direction without contamination (right panel), OS_θ sits slightly above F_θ (0.70 against 0.61 at $\delta = 0.6$), consistent with concentrating the same noncentrality on fewer degrees of freedom. The figures carry F_θ rather than OF_θ: within a size-adjusted pair the two are strictly increasing transforms of one another, so the rejection regions coincide identically and the curves are indistinguishable.

6 Illustration: what each statistic detects

6.1 A synthetic contrast

The size and power results read together on a single design that frames the empirical application of Section 6.2. The design turns on one distinction: an SLX effect aligned with β_{nc}, $(\theta_1, \theta_2) = \tau(\beta_1, \beta_2)$, is locally indistinguishable from a lag or a spatial error, whereas an orthogonal effect, $(\theta_1, \theta_2) = \tau(-\beta_2, \beta_1)$ with $\beta_{nc} = (1, 2)$, cannot be reproduced by either, and that orthogonal direction is exactly what OS_θ isolates (Figure 3).

Five data-generating processes are run on the common hexagonal lattice at $n = 100$, with the same regressors x_1, x_2 and the same $\varepsilon \sim \mathcal{N}(0, I)$ across cells, each a restriction of the

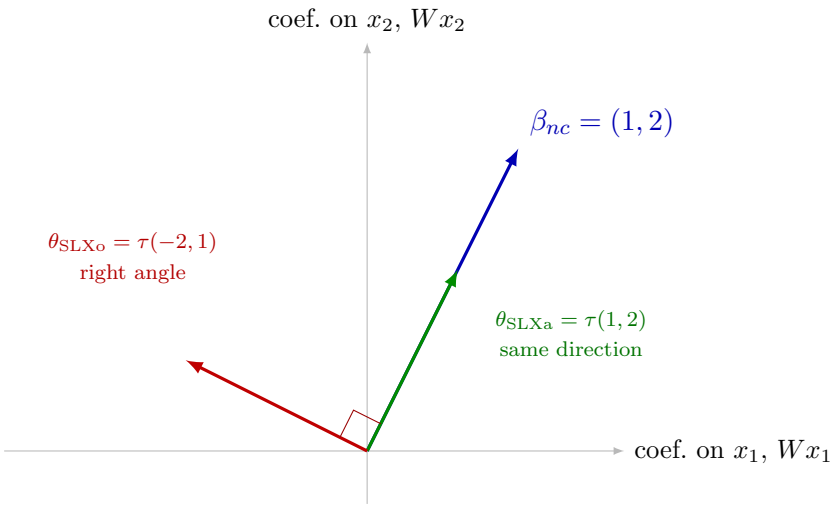


Figure 3: The two SLX coefficient directions. The aligned effect θ_{SLXa} is a positive multiple of β_{nc}; the orthogonal effect θ_{SLXo} meets it at a right angle (inner product zero). Only the orthogonal direction lies outside what a spatial lag or error can mimic.

encompassing model (4) obtained by activating one channel at a time:

$$\begin{aligned}
\text{null:} \quad & y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon, \\
\text{SLM:} \quad & y = \rho\, Wy + \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon, \\
\text{SEM:} \quad & y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, \quad u = \lambda\, Wu + \varepsilon, \\
\text{SLXa:} \quad & y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \tau\,(Wx_1) + 2\tau\,(Wx_2) + \varepsilon, \\
\text{SLXo:} \quad & y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 - 2\tau\,(Wx_1) + \tau\,(Wx_2) + \varepsilon,
\end{aligned} \tag{17}$$

where $\rho = \lambda = \tau = 0.3$ set the spatial intensity, ρ in the SLM cell, λ in the SEM cell and τ scaling the lagged effect in the two SLX cells, with $(\beta_0, \beta_1, \beta_2) = (1, 1, 2)$. The lagged coefficients $(\tau, 2\tau)$ in SLXa reproduce the covariate pattern $(1, 2)$, while $(-2\tau, \tau)$ in SLXo do not, which is the aligned-versus-orthogonal contrast of Figure 3 in the numbers themselves. With two non-constant regressors the SLX block contributes two coefficients; in the single-regressor case $k = 2$ it reduces to one, where OS_θ is no longer defined. Table 4 reports rejection at nominal 5% for the full battery, all calibrated against asymptotic critical values.

Table 4: Rejection rate (%) at nominal 5% under the five processes of (17), hexagonal W, $n = 100$, 5000 replications, common random numbers across columns. PS_θ is numerically RS^*_ρ and RS^*_λ is the Anselin et al. (1996) robust error test.

DGP	RS_ρ	RS_λ	PS_θ (= RS^*_ρ)	RS^*_λ	RS_θ	OS_θ	OF_θ
null	5.2	4.6	5.5	4.9	5.2	5.3	5.0
SAR	**93.7**	**61.0**	**85.9**	5.3	**80.2**	**6.1**	**5.7**
SLXa	67.5	5.9	82.3	26.2	73.5	4.3	4.0
SLXo	6.7	6.1	6.7	5.9	73.7	81.1	80.4
SEM	**23.0**	**47.0**	**9.0**	**35.4**	**9.9**	**8.4**	**8.0**

Three readings sit in the one table. The SLXo row is the positive statement: an SLX process is present, yet the entire classical battery reads it as no spatial dependence at all, 6.7%, 6.1%, 6.7% and 5.9% for RS_ρ, RS_λ, RS^*_ρ and RS^*_λ, every one within Monte Carlo error of nominal. In this design, OS_θ is the only statistic in the battery with substantial power against the orthogonal SLX alternative, with 81%, and it does so with more power than RS_θ's 74% because it concentrates the same noncentrality on $k - 2$ degrees of freedom rather than $k - 1$.

No statistic already in the sequence fires in that row, which is what makes the SLX branch's content irreducible rather than a refinement of an existing test. The SLM and SLXa rows are the concession: there the battery is loud in both and loud alike, RS^*_ρ at 85.9 and 82.3 and RS_θ at 80.2 and 73.5, so a rejection cannot be attributed to one specification over the other, and OS_θ is correctly silent in both, 6.1% and 4.3%, because along the common-factor direction the lag, the error and the aligned SLX effect are locally observationally equivalent (Proposition 4). The SEM row carries the known vulnerability in plain sight: OS_θ over-rejects at 8.4%, the error-parameter distortion of Section 5.2, correctable in closed form by the rescaling of (16). The empirical application of the Section 6.2 is an instance of the SLM and SLXa rows on real data: a battery that rejects and an OS_θ that does not.

6.2 Empirical application: covariate-specific spillovers across five datasets

Beyond a single case, we apply the complete testing sequence to five standard public cross-sections. Together, these datasets cover the regimes considered in the synthetic design of Section 6.1. The applications are:

1. The growth data of Ertur and Koch (2007) ($n = 91$ countries, $k = 3$), relating log per-capita income in 1995 to the log investment rate and $\ln(n + g + \delta)$.

2. The Columbus crime data of Anselin (1988) ($n = 49$, $k = 3$), relating the residential-crime rate to household income (INC) and housing value (HOVAL).

3. The St. Louis county homicide data ($n = 78$, $k = 3$), relating the homicide rate to a resource-deprivation index (RDAC90) and police expenditure (PE87).

4. The Baltimore house-sale data ($n = 211$, $k = 10$), relating sale prices to nine hedonic characteristics: rooms, bathrooms, patio, fireplace, air conditioning, garage, age, lot size, and finished area.

5. The Georgia county educational-attainment data ($n = 159$, $k = 6$), relating the share of adults with a bachelor's degree to the shares of the population that are rural, elderly, foreign-born, in poverty, and Black.

The weight matrix is the one bundled with each source. For Ertur–Koch, it is the row-standardized inverse-squared great-circle distance matrix between national capitals, corresponding to the first form of their equation (22). For the other four datasets, we use the row-standardized first-order contiguity matrix distributed with each libpysal example. These matrices define symmetric binary neighbor graphs, with equal within-row weights and approximately five neighbors per unit on average: 4.8 for Columbus, 5.1 for St. Louis, 5.6 for Baltimore, and 5.4 for Georgia. All datasets have $k \geq 3$, so that OS_θ is defined, and none is a Monte Carlo construction.

Table 5 reports the score statistics for each dataset: the simple RS_ρ and RS_λ tests, the robust RS^*_ρ and RS^*_λ tests of Anselin et al. (1996), and the SLX score tests RS_θ and $RS^*_{\theta|\rho}$ of Koley and Bera (2024). It also provides the corresponding statistics from the synthetic design in Section 6.1 and from the testing sequence developed below. Table 6 reports the common-factor component PS_θ (numerically equal to RS^*_ρ), the orthogonal remainder OS_θ, its exact-F and rescaled-sandwich p-values, the Breusch–Pagan diagnostic, and the SARAR estimates $(\hat{\rho}, \hat{\lambda})$, which locate each dataset relative to the error-contamination case examined in Section 5.2.[3]

All five applications are computed from OLS using the bundled weight matrix. The additive identity $RS_\theta = PS_\theta + OS_\theta$ holds to eleven decimal places in every row. The reported results are also reproduced by the R package `slxos` (see Declarations), a companion implementation of the full testing battery.

Table 5: The score battery on the same five cross-sections. The first four are the Anselin et al. (1996) LM tests: the simple lag RS_ρ and error RS_λ, and their robust counterparts RS^*_ρ, RS^*_λ (χ^2_1 each). RS_θ and $RS^*_{\theta|\rho}$ are the Koley and Bera (2024) SLX score test and its version adjusted against ρ alone (χ^2_{k-1}). Each cell is the statistic with the p-value in parentheses, in **bold** when significant at 5%. RS^*_ρ reappears as PS_θ, and $RS_\theta = PS_\theta + OS_\theta$, $RS^*_{\theta|\rho} = RS^*_\lambda + OS_\theta$, in Table 6.

			Anselin et al. (1996) (χ^2_1)				Koley and Bera (2024) (χ^2_{k-1})	
Dataset	n	k	RS_ρ	RS_λ	RS^*_ρ	RS^*_λ	RS_θ	$RS^*_{\theta\|\rho}$
Ertur–Koch	91	3	**51.21** (<0.01)	**38.59** (<0.01)	**15.25** (<0.01)	2.63 (0.10)	**15.97** (<0.01)	3.35 (0.19)
Columbus	49	3	**8.90** (<0.01)	**5.21** (0.02)	**3.74** (0.05)	0.04 (0.83)	**6.36** (0.04)	2.67 (0.26)
Georgia	159	6	**5.20** (0.02)	**4.18** (0.04)	1.19 (0.28)	0.18 (0.67)	2.53 (0.77)	1.51 (0.91)
St. Louis	78	3	**9.36** (<0.01)	**10.91** (<0.01)	0.08 (0.78)	1.63 (0.20)	**13.57** (<0.01)	**15.12** (<0.01)
Baltimore	211	10	**45.44** (<0.01)	**8.20** (<0.01)	**40.58** (<0.01)	3.33 (0.07)	**68.34** (<0.01)	**31.10** (<0.01)

Table 6: The decomposition on the same five cross-sections. $PS_\theta = RS^*_\rho$ is the Anselin et al. (1996) robust lag test of Table 5 (common-factor direction); OS_θ is the orthogonal remainder on $k-2$ degrees of freedom; OF_θ is its exact-F p-value under homoskedastic normality; swλ is the rescaled-sandwich p-value of Section 5.3, robust to spatial-error dependence, from the single SARAR fit $(\hat\rho, \hat\lambda)$ in the last column; BP is the studentized Breusch–Pagan p-value of Koenker (1981), a heteroskedasticity diagnostic.

Dataset	n	k	PS_θ	OS_θ	OF_θ (p)	swλ (p)	BP (p)	$(\hat\rho, \hat\lambda)$
Ertur–Koch	91	3	15.25	0.72	0.41	0.48	0.07	$(+0.48, +0.40)$
Columbus	49	3	3.74	2.62	0.12	0.13	0.03	$(+0.37, +0.15)$
Georgia	159	6	1.19	1.34	0.87	0.87	<0.001	$(+0.15, +0.11)$
St. Louis	78	3	0.08	**13.49**	<0.001	0.01	<0.001	$(+0.01, +0.71)$
Baltimore	211	10	40.58	**27.76**	<0.001	<0.001	<0.001	$(+0.54, -0.47)$

Table 5 reveals substantial heterogeneity in the spatial evidence across the five cross-sections. The Ertur–Koch and Baltimore applications exhibit strong evidence against the null of no spatial dependence, with the robust spatial-lag statistic RS^*_ρ reaching 15.25 and 40.58, respectively, both significant at the 1% level. The Columbus data provide evidence of spatial dependence through the robust spatial-lag statistic, $RS^*_\rho = 3.74$ ($p = 0.05$), while the robust error statistic is clearly nonsignificant, $RS^*_\lambda = 0.04$ ($p = 0.83$). In contrast, the Georgia data illustrate a case in which both simple LM statistics are significant, $RS_\rho = 5.20$ and $RS_\lambda = 4.18$, but neither robust counterpart rejects: $RS^*_\rho = 1.19$ ($p = 0.28$) and $RS^*_\lambda = 0.18$ ($p = 0.67$). St. Louis shows a similar attenuation after robustification, with neither $RS^*_\rho = 0.08$ ($p = 0.78$) nor $RS^*_\lambda = 1.63$ ($p = 0.20$) being significant. The SLX score statistics also vary across applications. The unadjusted statistic RS_θ is significant in four datasets, except Georgia, and the statistic $RS^*_{\theta|\rho}$ is significant for St. Louis, and Baltimore, but not for Ertur–Koch, Columbus or Georgia. These contrasting patterns motivate the decomposition examined in the Table 6, which separates the common-factor component from the orthogonal covariate-specific component of the SLX score.

Table 6 shows that in three of the five applications in which the test battery rejects, the evidence is confined to the common-factor direction, with no additional evidence from OS_θ against the common-factor restriction. Ertur–Koch provides the clearest example. The value of $PS_\theta = 15.25$ indicates strong evidence of a spatial effect, whereas $OS_\theta = 0.72$ ($p = 0.41$) does not reject the null hypothesis. Consistent with this result, the fitted SLX coefficient is nearly collinear with $\hat\beta_{nc}$, with $\cos(\hat\theta, \hat\beta_{nc}) = 0.98$ under the $\hat J_{\theta\theta.1}$ metric. Thus, the evidence associated with WX is consistent with the common-factor direction and does not reveal an additional covariate-specific component.

For the Columbus crime data, $PS_\theta = 3.74$ is significant to the conventional significance threshold ($p = 0.05$), whereas $OS_\theta = 2.62$ does not reject ($p = 0.12$). This suggests that any spatial signal detected by the common-factor component is not accompanied by statistically detectable covariate-specific spillover content. Georgia exhibits little evidence of spatial dependence: neither $PS_\theta = 1.19$ nor $OS_\theta = 1.34$ is statistically significant.

Overall, these three applications illustrate that the SLX decomposition does not identify a statistically significant component beyond the common-factor direction. Consequently, the additional information provided by OS_θ does not alter the conclusions obtained from the classical test statistics.

The other two datasets are where the decomposition earns its keep. St. Louis homicide is the empirical counterpart of the SLXo row of Table 4: the robust lag test is essentially zero, $PS_\theta = 0.08$ ($p = 0.78$), and the classical battery is silent in the common-factor direction, yet $OS_\theta = 13.49$ is large and $\cos(\hat{\theta}, \hat{\beta}_{nc}) = 0.08$, an SLX coefficient almost orthogonal to $\hat{\beta}_{nc}$. A practitioner consulting only Anselin et al. (1996)'s four statistics would read no lag and no error and stop, missing off-line content that lives entirely away from the common-factor line and that only OS_θ detects: the entire signal, $RS_\theta = 0.08 + 13.49$, is the part the classical battery has no coordinate for. It is the clearest of the five cases in which the $k-2$ residual carries the whole of the off-line signal; the strength of that signal, however, should be read against the calibration we trust rather than against the raw statistic. Two caveats bind that reading. First, St. Louis carries a high fitted error, $\hat{\lambda} = 0.71$. The pre-test RS^*_λ does not flag it ($p = 0.20$), so we do not read the exact-F ($p = 0.0002$) literally; the rescaled sandwich of Section 5.3, which is robust to that contamination, is the calibration we report (swλ $p = 0.015$, Table 6), so the off-line content is not an artifact of error dependence. Second, and not repaired by that correction, the studentized Breusch–Pagan statistic rejects homoskedasticity for St. Louis (column BP, Table 6), and swλ rescales the variance for Ω_λ but not for heteroskedasticity, so its $p = 0.015$ still rests on a homoskedastic error variance. The off-line reading is therefore a rejection at the 5% level under a calibration that corrects one of the two contaminations present, and a heteroskedasticity-robust OS_θ (Section 7) is what would settle it.

Baltimore house prices show both channels large at once: $PS_\theta = 40.58$ in the common-factor direction and $OS_\theta = 27.76$ off it. The rejection says the lagged regressors do not enter in proportion to their own coefficients, so a single aggregated column $WX_{nc}\beta_{nc}$ cannot represent them, a pattern consistent with heterogeneous, covariate-specific spillovers; as an omnibus test over the orthogonal directions, however, OS_θ does not identify which of the hedonic characteristics carries the off-line content (Remark 2). Baltimore is also the case that exposes the fragility of the calibration rule itself, though not for the reason $\hat{\lambda}$'s magnitude might suggest. The pre-test clears the 0.05 gate only marginally, RS^*_λ $p = 0.07$, and Step 3's rule keys the choice between OF_θ and the rescaled sandwich exclusively to this single threshold: a small perturbation of the data could have placed Baltimore on the other side of the gate. Yet across all five datasets, including St. Louis and Baltimore, where $\hat{\lambda}$ is largest in magnitude, 0.71 and -0.47, the sandwich correction never overturns the conclusion reached by OF_θ ($p = 0.0005$ versus $p = 0.0001$ here).

Read across the five, the decomposition separates two situations that the classical battery reports identically. Where OS_θ is null (Ertur–Koch, Columbus, Georgia), a significant robust-lag or Durbin reading is confined to the common-factor direction and cannot be attributed to spatially lagged regressors specifically; where OS_θ rejects (St. Louis, Baltimore) there is spillover content orthogonal to that direction, and St. Louis is the extreme in which the common-factor projection is empty and the whole signal is invisible without OS_θ. The point for applied work is that the same four classical numbers are compatible with both regimes, and only the $k-2$ residual tells them apart.

These readings are bounded by considerations that apply across all five datasets, not to any one of them in particular. The cross-sections are heterogeneous so the finite-sample exactness of OF_θ is nominal rather than literal and the rescaled sandwich of Section 5.3 is the reference

throughout (Assumption 2, Section 7). This is not left to assertion: the studentized Breusch–Pagan test of Koenker (1981) on the OLS residuals (column BP of Table 6), robust to non-normality, does not reject homoskedasticity for Ertur–Koch at conventional levels ($p = 0.07$) but rejects it for Columbus, Georgia, St. Louis, and Baltimore ($p < 0.05$ in all four); on those four, accordingly, the rescaled sandwich swλ is the calibration we rely on, with the qualification that swλ corrects for the spatial-error channel and not for heteroskedasticity itself, so on those four even that calibration is nominal rather than literal until the heteroskedasticity-robust variant of Section 7 is available. A non-rejection of OS_θ bounds the SLX contribution to the common-factor line and does not establish $\theta = 0$ (Remark 1). What the five cases establish is narrower and, for the purpose of the decomposition, sufficient: that the increment OS_θ over the classical battery is empirically neither always null nor always informative, but a coordinate that in real data ranges from redundant to decisive.

Reading the classical battery through the decomposition

The steps below do not append a new battery; they read a single OLS fit in which the classical battery is already contained. The one quantity the SLX branch contributes beyond the practitioner's table is OS_θ, the test of the $k-2$ directions of θ orthogonal to the common-factor line β_{nc}.

1. Run OLS and compute the standard battery of Anselin et al. (1996): RS_ρ, RS_λ, RS^*_ρ, RS^*_λ (Table 5 for the five datasets). If $k = 2$, stop: Corollary 1 says the SLX branch adds nothing.

2. If $k \geq 3$, compute RS_θ, OS_θ and the decomposition $RS_\theta = PS_\theta + OS_\theta$. Interpret PS_θ as evidence concerning the common-factor component $\hat{\beta}'_{nc}\theta$, rather than as evidence uniquely attributable to a spatially lagged dependent variable.

3. Assess the potential impact of spatial-error dependence. Use RS^*_λ as a diagnostic for whether the asymptotic/exact calibration of OS_θ may be affected. If not significant, OF_θ provides the exact finite-sample counterpart under normality. When it rejects, fit a single SARAR model under $H_0 : \theta = 0$ and report the rescaled-sandwich statistic of Section 5.3 for OS_θ instead. This diagnostic should not be interpreted as a reliable classifier of the underlying DGP, particularly under large fixed values of ρ.

4. Interpret OS_θ as a test of whether the SLX branch carries information orthogonal to the common-factor direction. A large value is evidence for heterogeneous spatial effects across covariates; a non-rejection does not imply $\theta = 0$, it is consistent with potentially large effects satisfying $\theta = \tau\beta_{nc}$. Symmetrically, a rejection of the common-factor restriction with a non-significant OS_θ points to Wy, not to WX: a pure lag with $\theta = 0$ produces exactly that pattern, so only OS_θ, which measures the off-line content, would make the Durbin term unambiguous.

Symmetrically, two significant robust tests RS^*_ρ and RS^*_λ with a non-significant OS_θ do not establish a SARAR process: the common-factor direction is then saturated, and an SLX effect aligned with β_{nc} reproduces the same pair of rejections with $\rho = \lambda = 0$, so separating lag, error, and aligned SLX in that configuration requires the fitted model rather than the OLS scores.

OS_θ therefore belongs at the level of the simple Lagrange multiplier tests, not as a stage conditioned on them: it is orthogonal by construction to the direction carrying the first-order signature of both a lag and a common-factor error, so it reads the $k-2$ contrasts those tests cannot and is a complementary coordinate of the same OLS fit rather than a pretest selected by Step 1. The role of RS^*_λ in Step 3 is correspondingly narrow: it neither classifies the DGP nor changes the statistic, only flagging when the non-local λ effect of Section 5.2 threatens the calibration, at which point the rescaled sandwich of Section 5.3 supplies the reference.

Step 3's limitation is not hypothetical. It is a size phenomenon of the classical battery itself: under a fixed spatial lag the robust error test RS^*_λ need not retain nominal size, and where it fails a rejection of RS^*_λ cannot be read as evidence of error contamination. At $\rho = 0.3$ the distortion is absent, RS^*_λ rejecting at 5.3% in the SLM row of Table 4; at $\rho = 0.7$ and $\lambda = 0$, however, it rises with the sample size, so that a reading anchored on RS^*_λ would place the departure in the error term when the true process is a pure spatial lag. The channel is S_ρ, on which the Bera and Yoon (1993) adjustment defining RS^*_λ is built and which grows with the strength of the lag, so the distortion is worst precisely where a lag is strongest, and it is the classical statistic, and not OS_θ, that carries it.

This exercise is a negative result about the classical battery, not a stage of the proposed reading: it shows that RS^*_λ and its companions are unreliable as DGP classifiers under a strong fixed lag, which is exactly why OS_θ is offered as a coordinate of the OLS fit to be read rather than a pretest to condition on.

7 What remains open

The exact decomposition of Theorem 1, its finite-sample and asymptotic null laws, and the size behavior of OS_θ are settled; a few lines of inquiry remain open, though none affects the main result.

Along a drifting sequence $\beta_{nc,n} = n^{-1/2}b$ the asymptotic distribution of OS_θ is unresolved, since Theorem 2's route does not cover the rank discontinuity at $\beta_{nc} = 0$ that the framework of Bera et al. (2010) would need to accommodate; the finite-sample law, however, is complete and holds for every β_{nc} including zero (Proposition 5), so the decomposition and its exactness stand regardless. The effect of estimating the direction is likewise of second order: the conditional noncentrality at $\theta = \tau\beta_{nc}$ is $O_p(\tau^2)$, flat in n and numerically small, and although a fixed large ρ lets the estimated direction lower the size distortion by a few points relative to the population direction, an adaptation we trace to the unpurged higher-order terms of (15) without a full account, neither channel moves the readings of Section 6.2.

A second group does bear on applied use, and heteroskedasticity is the main entry. Since the exact Beta law and the closed-form λ correction both rest on homoskedastic normal errors, the reported inference should therefore be interpreted as asymptotically calibrated rather than literally exact wherever the Breusch–Pagan diagnostic of Section 6.2 rejects; a size check under covariate-driven heteroskedasticity (Supplementary Material) nonetheless keeps OS_θ close to nominal across a 90/10 standard-deviation ratio up to 13, an empirical robustness the homoskedastic theory does not itself guarantee.

The third group collects natural extensions beyond the homoskedastic-normal setting studied here. A heteroskedasticity-robust OS_θ, whether it retains any exact finite-sample law or only an asymptotic one, is the priority for cross-sectional data. The linear structure of the score makes a first version concrete. Since $\hat{P}'S_\theta = Z'\hat{e}/\hat{\sigma}^2$ with $Z = WX_{nc}\hat{P}$ has as its only random ingredient the linear form $Z'\hat{e} = (MZ)'e$, under independent but heteroskedastic errors $\mathrm{Var}(Z'\hat{e}) = (MZ)'\Sigma(MZ)$ with $\Sigma = \mathrm{diag}(\sigma_i^2)$, estimated consistently by the White matrix $\widehat{V} = \sum_i \hat{e}_i^2\, \tilde{z}_i\tilde{z}_i'$, $\tilde{z}_i$ the i-th row of MZ; the statistic $OS^{\mathrm{HC}}_\theta = (Z'\hat{e})'\widehat{V}^{-1}(Z'\hat{e})$ is then χ^2_{k-2} under H_0^{OS} and heteroskedasticity of unknown form, a linear form to which a Lindeberg argument applies with no quadratic term to control, at the cost of the exact Beta law of Proposition 5, which rests on an isotropy that heteroskedasticity destroys. This handles heteroskedasticity but not spatial-error dependence, under which $\mathrm{Var}(Z'\hat{e}) = \sigma^2(MZ)'\Omega_\lambda(MZ)$ is not diagonal. The spatial HAC estimator of Kelejian and Prucha (2007) together with the heteroskedasticity-robust GM of Kelejian and Prucha (2010) provides a route toward a sandwich estimator robust to spatial dependence and heteroskedasticity at once, at the cost of the exactness that motivates OF_θ.

A separate open question concerns model-selection strategies. The specific-to-general se-

quence of Anselin et al. (1996) and the general-to-specific strategy of Florax et al. (2003) extract the same scalar combination from opposite ends through the decomposition $RS_\theta = PS_\theta + OS_\theta$. Whether this symmetry has implications for their relative power in the presence of WX terms is a question that Theorem 1 opens but does not close.

Declarations

Disclosure statement. The author reports there are no competing interests to declare.

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability statement. The five cross-sections used in Section 6.2 are publicly available example datasets distributed with the spData/spreg and libpysal packages, cited at first use in the text; no new data were collected for this study.

Code availability. All statistics, the Monte Carlo experiments of Section 5, and the empirical application of Section 6.2 are implemented in the R package `slxos`, available at https://github.com/mherreragomez/slxos and archived at a permanent, citable DOI: https://doi.org/10.5281/zenodo.22836693. Replication scripts for every table and figure in this paper are included in the package's `inst/replication/` directory.

Generative AI use. Generative AI tools (a large language model) were used extensively throughout the preparation of this manuscript and of the accompanying R package `slxos`, including drafting and revising text, exploring and structuring arguments, assisting in the verification of algebraic derivations and proofs, and assisting in the development, debugging, and testing of the code. All research design choices, the mathematical proofs, the statistical results, and the interpretation of the empirical findings are the author's own. The author reviewed all AI-assisted suggestions, independently verified every derivation, proof, and numerical result reported in the paper, made all substantive scientific decisions, and takes full responsibility for the final manuscript and code.

References

Anselin, L. (1988). Lagrange multiplier test diagnostics for spatial dependence and spatial heterogeneity. *Geographical Analysis*, 20(1):1–17.

Anselin, L., Bera, A. K., Florax, R., and Yoon, M. J. (1996). Simple diagnostic tests for spatial dependence. *Regional Science and Urban Economics*, 26(1):77–104.

Anselin, L. and Florax, R. J. G. M. (1995). Small sample properties of tests for spatial dependence in regression models: Some further results. In Anselin, L. and Florax, R. J. G. M., editors, *New Directions in Spatial Econometrics*, pages 21–74. Springer, Berlin, Heidelberg.

Anselin, L., Serenini, R., and Amaral, P. (2026). Alternative perspectives on spatial econometrics model specification search. *Spatial Economic Analysis*, 21(2):288–335.

Baltagi, B. H. and Li, Q. (1991). A joint test for serial correlation and random individual effects. *Statistics & Probability Letters*, 11(3):277–280.

Baltagi, B. H. and Li, Q. (1995). Testing AR(1) against MA(1) disturbances in an error component model. *Journal of Econometrics*, 68(1):133–151.

Bera, A. K., Montes-Rojas, G., and Sosa-Escudero, W. (2010). General specification testing with locally misspecified models. *Econometric Theory*, 26(6):1838–1845.

Bera, A. K., Sosa-Escudero, W., and Yoon, M. (2001). Tests for the error component model in the presence of local misspecification. *Journal of Econometrics*, 101(1):1–23.

Bera, A. K. and Yoon, M. J. (1993). Specification testing with locally misspecified alternatives. *Econometric Theory*, 9(4):649–658.

Burridge, P. (1981). Testing for a common factor in a spatial autoregression model. *Environment and Planning A*, 13(7):795–800.

Cliff, A. D. and Ord, J. K. (1981). *Spatial Processes: Models and Applications*. Pion, London.

Ertur, C. and Koch, W. (2007). Growth, technological interdependence and spatial externalities: theory and evidence. *Journal of applied econometrics*, 22(6):1033–1062.

Florax, R. and Folmer, H. (1992). Specification and estimation of spatial linear regression models: Monte carlo evaluation of pre-test estimators. *Regional Science and Urban Economics*, 22(3):405–432.

Florax, R. J. G. M., Folmer, H., and Rey, S. J. (2003). Specification searches in spatial econometrics: the relevance of Hendry's methodology. *Regional Science and Urban Economics*, 33(5):557–579.

Gibbons, S. and Overman, H. G. (2012). Mostly pointless spatial econometrics? *Journal of Regional Science*, 52(2):172–191.

Godfrey, L. G. and Veall, M. R. (2000). Alternative approaches to testing by variable addition. *Econometric Reviews*, 19(2):241–261.

Halleck Vega, S. and Elhorst, J. P. (2015). The SLX model. *Journal of Regional Science*, 55(3):339–363.

Kelejian, H. H. and Prucha, I. R. (1998). A generalized spatial two-stage least squares procedure for estimating a spatial autoregressive model with autoregressive disturbances. *Journal of Real Estate Finance and Economics*, 17(1):99–121.

Kelejian, H. H. and Prucha, I. R. (1999). A generalized moments estimator for the autoregressive parameter in a spatial model. *International Economic Review*, 40(2):509–533.

Kelejian, H. H. and Prucha, I. R. (2001). On the asymptotic distribution of the Moran I test statistic with applications. *Journal of Econometrics*, 104(2):219–257.

Kelejian, H. H. and Prucha, I. R. (2007). HAC estimation in a spatial framework. *Journal of Econometrics*, 140(1):131–154.

Kelejian, H. H. and Prucha, I. R. (2010). Specification and estimation of spatial autoregressive models with autoregressive and heteroskedastic disturbances. *Journal of Econometrics*, 157(1):53–67.

Koenker, R. (1981). A note on studentizing a test for heteroscedasticity. *Journal of Econometrics*, 17(1):107–112.

Koley, M. (2024). *Specification Testing of Spatial Econometric Models*. PhD thesis, University of Illinois at Urbana-Champaign. https://hdl.handle.net/2142/124329.

Koley, M. and Bera, A. K. (2024). To use, or not to use the spatial Durbin model? that is the question. *Spatial Economic Analysis*, 19(1):30–56.

Mur, J. and Angulo, A. (2006). The spatial Durbin model and the common factor tests. *Spatial Economic Analysis*, 1(2):207–226.

Mur, J. and Angulo, A. (2009). Model selection strategies in a spatial setting: Some additional results. *Regional Science and Urban Economics*, 39(2):200–213.

Appendix

This appendix collects the proofs deferred from the main text, together with brief summaries of two supplementary investigations referenced above.

A Notation

Table 7: Notation. Symbols local to a single proof are omitted. Assumptions 1–3, stated in Section 2, give the regularity conditions on W, e, and X.

Parameters	
ρ	coefficient on the spatial lag Wy
λ	coefficient on the spatial error Wu
θ	$(k-1)$-vector of coefficients on WX_{nc}
β	full regression coefficient vector, $\beta = (\beta_1, \beta_{nc}')'$
β_{nc}	non-constant part of β, the coefficients on X_{nc}; population value, used only in the information blocks, local noncentralities and asymptotics
$\hat{\beta}_{nc}$	OLS estimate of β_{nc}; the vector appearing in every computed statistic and in every in-sample equality (Section 3)
σ^2, $\hat{\sigma}^2$	error variance; $\hat{\sigma}^2 = \hat{e}'\hat{e}/n$ replaces it in every computed statistic
$\tilde{\sigma}^2$	rescaled variance estimate $\hat{e}'\hat{e}/\operatorname{tr}(M\hat{\Omega}_\lambda)$, (16); used by sw$\lambda$
$\hat{\rho}, \hat{\lambda}$	SARAR fit under $H_0 : \theta = 0$; $\hat{\lambda}$ builds $\hat{\Omega}_\lambda$ (Section 5.3)
Dimensions	
n	sample size
k	number of columns of X, including the intercept
$k-1$	non-constant columns of X; dimension of θ and β_{nc}
$k-2$	degrees of freedom of OS_θ for $k \geq 3$ (Proposition 4)
Matrices and information blocks	
W	$n \times n$ row-standardized weights matrix, common to lag and error
X, X_{nc}	$X = [\iota, X_{nc}]$, the full and non-constant regressor matrices
$\hat{e}$	OLS residual $\hat{e} = My$; $\hat{\sigma}^2 = \hat{e}'\hat{e}/n$
M	$I - X(X'X)^{-1}X'$, the OLS annihilator
Ω_λ, $\hat{\Omega}_\lambda$	$(I-\lambda W)^{-1}(I-\lambda W')^{-1}$ and its plug-in at $\hat{\lambda}$; the error covariance factor of the rescaled sandwich (Section 5.3)
$J_{\theta\theta.1}$	information block of θ partialled on (β, σ^2), (6); population object, built from σ^2
$\hat{J}_{\theta\theta.1}$	$X_{nc}'W'MWX_{nc}/\hat{\sigma}^2$, the same block at the OLS fit; the one that appears in (8), (9), (10) and (12)
$J_{\rho\rho.1}$, $\hat{J}_{\rho\rho.1}$	$\beta_{nc}'J_{\theta\theta.1}\beta_{nc} + T_W$ and its plug-in $\hat{\beta}_{nc}'\hat{J}_{\theta\theta.1}\hat{\beta}_{nc} + T_W$
T_W	$\operatorname{tr}(W'W + W^2)$, shared by $J_{\rho\rho.1}$ and $J_{\lambda\lambda.1}$; free of β_{nc} and σ^2, so it carries no accent
$\hat{J}^*_{\theta\theta.1}$	$\hat{J}_{\theta\theta.1}$ after the joint (ρ, λ) adjustment, (9); kernel span$(\hat{\beta}_{nc})$, rank $k-2$ in every sample. Its population counterpart $J^*_{\theta\theta.1}$ has rank $k-1$ at $\beta_{nc} = 0$
$\hat{P}$	$(k-1) \times (k-2)$ matrix spanning the $\hat{J}_{\theta\theta.1}$-orthogonal complement of $\hat{\beta}_{nc}$ (Proposition 4); not the subspace containing $\tilde{S}_\theta$, which is the ordinary complement of $\hat{\beta}_{nc}$
P	its population counterpart, the $J_{\theta\theta.1}$-orthogonal complement of β_{nc}; appears only in Remark 1 and Theorem 2
Scores	

continued on next page

Table 7 (continued)

S_θ	SLX score $X'_{nc}W'\hat{e}/\hat{\sigma}^2$ at the OLS fit, (5)
S_ρ	lag score $\hat{e}'Wy/\hat{\sigma}^2$
S_λ	error score $\hat{e}'W\hat{e}/\hat{\sigma}^2$; the identity $S_\rho = S_\lambda + \hat{\beta}'_{nc}S_\theta$ holds in every sample (Proposition 1)
$\tilde{S}_\theta$	S_θ after removing its $\hat{\beta}_{nc}$ component, (8); OS_θ is its quadratic form in the $(\hat{J}^*_{\theta\theta.1})^+$ metric
$\tilde{S}_\rho$, $\tilde{S}_\lambda$	the analogous adjustments of S_ρ, S_λ; identically 0 in every sample (Proposition 3)
Statistics	
RS_θ	unadjusted SLX score test, $k-1$ degrees of freedom
PS_θ	projected score, $\equiv RS^*_\rho$ of Anselin et al. (1996), (12)
RS^*_ρ, RS^*_λ	robust lag and error tests of Anselin et al. (1996)
OS_θ	orthogonalized SLX test, $k-2$ degrees of freedom for $k \geq 3$
OF_θ	F test excluding $WX_{nc}\hat{P}$ only, $k-2$ restrictions; exact counterpart of OS_θ under normality (Section 4.1)
swλ	rescaled-sandwich version of OS_θ, robust to spatial-error dependence, χ^2_{k-2}; central matrix in $\hat{\Omega}_\lambda$, scale $\tilde{\sigma}^2$ (Section 5.3)
F_θ	F test excluding all of WX_{nc}, $k-1$ restrictions; finite-sample twin of RS_θ, not robust to ρ
$RS^*_{\theta\|\rho}$	singly robust SLX test, adjusted against ρ alone
$RS_{(\rho,\lambda)}$	joint SARAR score test
$\widetilde{RS}_\rho$, $\widetilde{RS}_\lambda$	the 0/0 statistics $\tilde{S}_\rho$, $\tilde{S}_\lambda$ would define; do not exist
BP	studentized Breusch–Pagan test of Koenker (1981), χ^2_{k-1}; heteroskedasticity diagnostic reported in Table 6

B Proofs deferred from Sections 3 and 4

Proof of Proposition 3. Write $J \equiv \hat{J}_{\theta\theta.1}$, $b \equiv \hat{\beta}_{nc}$, $\hat{D} \equiv b'Jb + T_W = \hat{J}_{\rho\rho.1}$. The shorthand b denotes the OLS estimate throughout this proof and the two that follow, since every equality established in them is an equality between computed quantities; the algebra is valid for any fixed nonzero vector, and it is Proposition 1 that pins the vector to $\hat{\beta}_{nc}$. The blocks inverted below are the plug-in ones, the two that involve neither β_{nc} nor σ^2, namely $J_{\theta\lambda.1} = 0$ and $J_{\lambda\lambda.1} = J_{\rho\lambda.1} = T_W$, being the same object either way. The first two cases share the same mechanics: partial out a 2×2 information block and read off the resulting coefficients. For $\tilde{S}_\rho$, conditioning on (λ, θ), block diagonal since $J_{\theta\lambda.1} = 0$, gives coefficients 1 on S_λ and b' on S_θ, so $\tilde{S}_\rho = S_\rho - S_\lambda - b'S_\theta = 0$ by Proposition 1. For $\tilde{S}_\lambda$, conditioning on (ρ, θ) analogously gives coefficients $(1, -b')$ from $\begin{pmatrix} \hat{D} & b'J \\ Jb & J \end{pmatrix}$ against $(T_W, 0')'$, so $\tilde{S}_\lambda = S_\lambda - S_\rho + b'S_\theta = 0$, again by Proposition 1.

For $\tilde{S}_\theta$, the conditioning block is (ρ, λ) with information $\begin{pmatrix} \hat{D} & T_W \\ T_W & T_W \end{pmatrix}$, whose determinant is $(b'Jb)T_W > 0$ whenever $b \neq 0$, so it is invertible; since $J_{\theta\lambda.1} = 0$, only the ρ-row of $J_{\theta,(\rho\lambda).1}$ is nonzero, but both entries of that row of the inverse enter the correction: the adjustment is $J_{\theta\rho.1}\big([J^{-1}_{(\rho\lambda).1}]_{11}S_\rho + [J^{-1}_{(\rho\lambda).1}]_{12}S_\lambda\big)$ with $[J^{-1}_{(\rho\lambda).1}]_{11} = T_W/\det = (b'Jb)^{-1}$ and $[J^{-1}_{(\rho\lambda).1}]_{12} = -T_W/\det = -(b'Jb)^{-1}$, and $J_{\theta\rho.1} = Jb$, so the correction term is exactly $Jb(S_\rho - S_\lambda)/(b'Jb)$, not $Jb\,S_\rho/(b'Jb)$ alone. Substituting $S_\rho - S_\lambda = b'S_\theta$ (Proposition 1) leaves the second expression in (8). For the subspace claim, write $\Pi = Jb\,b'/(b'Jb)$, so that $\tilde{S}_\theta = (I - \Pi)S_\theta$. The matrix Π is idempotent, with range $\mathrm{span}(Jb)$ and kernel $\{v : b'v = 0\}$: it is the oblique projector onto $\mathrm{span}(Jb)$ along the ordinary orthogonal complement of b, not the J-orthogonal projector onto $\mathrm{span}(b)$. Hence $\tilde{S}_\theta \in \mathrm{range}(I - \Pi) = \ker \Pi = \{v : b'v = 0\}$, which is the assertion $b'\tilde{S}_\theta = 0$; the direct verification is $b'\tilde{S}_\theta = b'S_\theta - (b'Jb)(b'S_\theta)/(b'Jb) = 0$. Note that $b'J\tilde{S}_\theta$ is not zero in general, so $\tilde{S}_\theta$ does *not* lie in the J-orthogonal complement of b; the J-orthogonality visible in Figure 1

is a statement in the whitened coordinates $u = J^{-1/2}S_\theta$, $v = J^{1/2}b$, in which $\tilde{S}_\theta = J^{1/2}(u - P_v u)$ and $b'\tilde{S}_\theta = v'(I - P_v)u = 0$. Both subspaces have dimension $k-2$ and both are $\{0\}$ iff $k-1 = 1$, which gives the stated equivalence with $k = 2$. Finally, $\tilde{S}_\theta = 0$ iff $S_\theta \in \ker(I - \Pi) = \text{span}(Jb)$, a set of Lebesgue measure zero for $k \geq 3$, which is the sense in which the third expression fails to vanish. □

Proof of Proposition 4. The correction term is computed as in the proof of Proposition 3; $\hat{J}^*_{\theta\theta.1}\hat{\beta}_{nc} = 0$ is immediate and the subtracted term has rank one. The population counterpart is the same expression with β_{nc} and $J_{\theta\theta.1}$ in place of $\hat{\beta}_{nc}$ and $\hat{J}_{\theta\theta.1}$, and $J^*_{\theta\theta.1}\beta_{nc} = 0$ by the identical algebra. For (10), note $\hat{P}'\hat{J}^*_{\theta\theta.1}\hat{P} = \hat{P}'\hat{J}_{\theta\theta.1}\hat{P}$ because $\hat{P}'\hat{J}_{\theta\theta.1}\hat{\beta}_{nc} = 0$ by construction, and $\hat{\beta}'_{nc}\tilde{S}_\theta = 0$ by (8), so $\tilde{S}_\theta$ lies in the range of $\hat{J}^*_{\theta\theta.1}$ and the pseudoinverse quadratic form coincides with the ordinary one in the $\hat{P}$ coordinates. Invariance to $\hat{P}$ follows because $(\hat{J}^*)^+$ is the $\hat{J}_{\theta\theta.1}$-metric inverse restricted to the complement. For the vanishing claim, (10) is zero if and only if $\hat{P}'S_\theta = 0$, and since the columns of $\hat{P}$ span $(\hat{J}_{\theta\theta.1}\hat{\beta}_{nc})^\perp$ this holds if and only if $S_\theta \in \text{span}(\hat{J}_{\theta\theta.1}\hat{\beta}_{nc})$, a line in $\mathbb{R}^{k-1}$ and hence a set of Lebesgue measure zero for $k \geq 3$. The corresponding population statement is Remark 1: $E[S_\theta] = J_{\theta\theta.1}\theta$ under a local alternative gives noncentrality zero if and only if $P'J_{\theta\theta.1}\theta = 0$, that is $\theta \in \text{span}(\beta_{nc})$. □

Proof of Theorem 1. Write $J \equiv \hat{J}_{\theta\theta.1}$, $b \equiv \hat{\beta}_{nc}$, $\hat{D} \equiv b'Jb + T_W$. For (12), $RS^*_\rho = (S_\rho - S_\lambda)^2/(\hat{J}_{\rho\rho.1} - T_W)$; the numerator is $(b'S_\theta)^2$ by Proposition 1 and the denominator is $b'Jb$ by (7).

For (13), set $u = J^{-1/2}S_\theta$ and $v = J^{1/2}b$, so that $RS_\theta = S'_\theta J^{-1}S_\theta = \|u\|^2$. The component of u along v has squared length $(v'u)^2/\|v\|^2 = (b'S_\theta)^2/(b'Jb) = PS_\theta$ by (12). In the same coordinates $\tilde{S}_\theta$ of (8) is $J^{1/2}(u - P_v u)$ with P_v the orthogonal projector on v, and $\hat{J}^*_{\theta\theta.1} = J^{1/2}(I - P_v)J^{1/2}$, so $OS_\theta = (\tilde{S}_\theta)'(\hat{J}^*)^+\tilde{S}_\theta = \|u - P_v u\|^2$. Pythagoras gives (13).

For (14), adjusting S_θ for ρ alone gives $S^\dagger = S_\theta - Jb\,S_\rho/\hat{D}$ and $J^\dagger = J - Jbb'J/\hat{D}$, so $RS^*_{\theta|\rho} = (S^\dagger)'(J^\dagger)^{-1}S^\dagger$. Substituting $S_\rho = S_\lambda + b'S_\theta$ throughout gives $b'S^\dagger = -\big(S_\lambda - (T_W/\hat{D})S_\rho\big)$ and $b'J^\dagger b = (b'Jb)T_W/\hat{D} = J_{\lambda\lambda.1} - T^2_W/\hat{D}$, so the projection of $S^\dagger$ onto b in the $J^\dagger$ metric has squared length $(b'S^\dagger)^2/(b'J^\dagger b) = RS^*_\lambda$, and the $J^\dagger$-orthogonal remainder reduces, substituting the definitions of $S^\dagger$ and $J^\dagger$, to $S_\theta - Jb(b'S_\theta)/(b'Jb)$, which is $\tilde{S}_\theta$ of (8). Pythagoras gives (14). □

Proof of Proposition 5. $\text{Cov}(X'y, Me) = \sigma^2 X'M = 0$ since $M = I - X(X'X)^{-1}X'$ annihilates X by construction, and $(X'y, Me)$ are jointly Gaussian, being linear in e; zero covariance under joint normality is independence. $\hat{\beta}_{nc}$, a function of $X'y$ alone, is therefore independent of Me, the vector every score in Proposition 1 is built from, and in particular independent of $\hat{P}$, which is a function of $\hat{\beta}_{nc}$ alone. Unconditionally $Me \sim \mathcal{N}(0, \sigma^2 M)$ and M is idempotent of rank $n-k$, so, conditional on the realized $\hat{P}$, Me is exactly isotropic within its own range. $Z = WX_{nc}\hat{P}$ itself need not lie in range(M), but the object the argument needs is its projection MZ, which is a fixed, rank-q subspace of range(M), independent of Me's direction within range(M). The projections of Me onto MZ and onto its orthogonal complement within range(M) are therefore independent central Gaussian vectors of dimension q and $n-k-q$, whatever $\hat{P}$ is, and B, the ratio of the first squared length to the total, is exactly Beta$(q/2, (n-k-q)/2)$ conditional on $\hat{P}$ and hence unconditionally, for every value of β_{nc}: the population value only ever enters through $\hat{P}$, which the argument has already conditioned out. Explicitly, with $Z = WX_{nc}\hat{P}$ and $\hat{e} = Me$, (10) reads $\hat{P}'S_\theta = Z'\hat{e}/\hat{\sigma}^2$ against $\hat{P}'\hat{J}_{\theta\theta.1}\hat{P} = Z'MZ/\hat{\sigma}^2$, so $OS_\theta = (Z'\hat{e})'(Z'MZ)^{-1}(Z'\hat{e})/\hat{\sigma}^2 = \hat{e}'P_{MZ}\hat{e}/\hat{\sigma}^2$ with $P_{MZ} = MZ(Z'MZ)^{-1}Z'M$ the orthogonal projector onto range(MZ); since $\hat{e}'P_{MZ}\hat{e} = SSR_r - SSR_u$ and $\hat{\sigma}^2 = SSR_r/n$, this is $OS_\theta = n(SSR_r - SSR_u)/SSR_r = nB$, the residual-sum-of-squares form quoted in the text. OF_θ is the textbook transform of the same B into an F ratio, exact for the same reason Godfrey and Veall (2000)-type variable-addition F tests are exact under normality. □

Proof of Lemma 1. For a quadratic form in *i.i.d.* errors, $\mathrm{Var}(e'We) = (\mu_4 - 3\sigma^4)\sum_i w_{ii}^2 + \sigma^4 \operatorname{tr}(W'W + W^2)$ and, for a linear form, $\mathrm{Cov}(b'e,\, e'We) = \mu_3 \sum_i b_i w_{ii}$; see Kelejian and Prucha (2001, Appendix A, Assumption 3) and Cliff and Ord (1981, Ch. 6). Both non-Gaussian contributions enter only through the diagonal of W, which is zero by construction. Hence $\mathrm{Var}(e'We) = \sigma^4 T_W$ and $\mathrm{Cov}(b'e, e'We) = 0$ whatever the error distribution. Since S_θ is linear in e and S_ρ, S_λ decompose into linear and quadratic parts by Proposition 1, every block of (6)–(7) is unaffected. □

C Proof of Theorem 2

Write $d_n = (S_\theta', S_\rho, S_\lambda)'$ for the score vector at the restricted MLE, $\hat{J}_n$ for the partitioned information evaluated at $(\hat\beta, \hat\sigma^2)$, and $\bar{J}_n$ for the same matrix evaluated at the true (β, σ^2). Let $\mathcal{J} = \lim_{n\to\infty} n^{-1}\bar{J}_n$, which exists, is finite, and has rank k by Assumption 3 together with Proposition 2. Throughout, $c_n = (-\hat\beta_{nc}', 1, -1)'$ and $c = (-\beta_{nc}', 1, -1)'$.

Step 1: limiting distribution of the score. Throughout this proof, e denotes the true error vector, not the OLS residual $\hat e$. Under H_0, $\hat e = Me$ because $M(X\beta + e) = Me$, so $S_\theta = (WX_{nc})'Me/\sigma^2 + o_p(1)$ is linear in e, $S_\lambda = e'MWMe/\sigma^2 + o_p(1)$ is purely quadratic, and $S_\rho = S_\lambda + \hat\beta_{nc}' S_\theta$ exactly by Proposition 1, with $\hat\beta_{nc} = \beta_{nc} + o_p(1)$ under Assumption 3, so that $S_\rho = S_\lambda + \beta_{nc}' S_\theta + o_p(n^{1/2})$.

Under Assumptions 1–3, the uniform boundedness of the row and column sums of W, the uniform boundedness of the elements of X, and the finite fourth moment of e are precisely the conditions of Theorem 1 of Kelejian and Prucha (2001), which gives

$$n^{-1/2} d_n \xrightarrow{d} \mathcal{N}(0, \mathcal{J}). \tag{18}$$

Under Lemma 1 the covariance in (18) is $\mathcal{J}$ whether or not e is Gaussian. Replacing (β, σ^2) by $(\hat\beta, \hat\sigma^2)$ affects d_n only at order $o_p(n^{1/2})$ because the information between the tested parameters and the nuisance parameters has been partialled out.

Step 2: the score lies in the column space of its own covariance. By Proposition 1, $c_n' d_n = -\hat\beta_{nc}' S_\theta + S_\rho - S_\lambda = 0$ identically, in every sample and for every n. Because $c_n' d_n = 0$ exactly, $c' n^{-1/2} d_n = (c - c_n)' n^{-1/2} d_n$ identically. Since $\hat\beta_{nc} \xrightarrow{p} \beta_{nc}$ under Assumption 3, $c_n \xrightarrow{p} c$, and therefore $(c - c_n)' n^{-1/2} d_n \xrightarrow{p} 0$ by Slutsky and (18). Hence the limiting normal vector in (18) satisfies $c'Z = 0$ almost surely, i.e. it is supported on $\mathcal{C}(\mathcal{J})$. This is the point at which the exactness of Proposition 1 matters: had the identity held only in expectation, $c_n' d_n$ would be $O_p(n^{1/2})$ rather than exactly zero, and the argument would fail.

Step 3: convergence of the projected weighting matrix. Fix a measurable selection $\hat P = P(\hat\beta_{nc})$, obtained for instance by $\hat J_{\theta\theta.1}$-orthogonalizing a fixed basis of $\mathbb{R}^{k-1}$ against $\hat\beta_{nc}$; by Proposition 4 the value of OS_θ does not depend on which selection is used. On the set $\beta_{nc} \neq 0$ the map $\beta_{nc} \mapsto P(\beta_{nc})$ is continuous, so $\hat\beta_{nc} \xrightarrow{p} \beta_{nc}$ gives $\hat P \xrightarrow{p} P$. Writing $\mathcal{J}_{\theta\theta} = \lim_n n^{-1} J_{\theta\theta.1}$, finite and positive definite by Assumption 3, the matrix $P'\mathcal{J}_{\theta\theta}P$ is positive definite of full rank $k-2$, and ordinary matrix inversion is continuous there, so $(n^{-1}\hat P'\hat J_{\theta\theta.1}\hat P)^{-1} \xrightarrow{p} (P'\mathcal{J}_{\theta\theta}P)^{-1}$. No generalized inverse appears at any point.

Step 4: conclusion. By Step 1 and Slutsky, $n^{-1/2}\hat P' S_\theta \xrightarrow{d} N(0,\, P'\mathcal{J}_{\theta\theta}P)$, a nondegenerate normal vector in $\mathbb{R}^{k-2}$. Combining with Step 3,

$$OS_\theta = \left(n^{-1/2}\hat P' S_\theta\right)' \left(n^{-1}\hat P'\hat J_{\theta\theta.1}\hat P\right)^{-1} \left(n^{-1/2}\hat P' S_\theta\right) \xrightarrow{d} \chi^2_{k-2},$$

the degrees of freedom being the dimension of the projected vector. Step 2 guarantees that the discarded direction carries no information asymptotically: the component of S_θ along $\hat{\beta}_{nc}$ is, by Proposition 1, exactly $S_\rho - S_\lambda$, and removing it is what distinguishes OS_θ from RS_θ. □

Remark on the drifting case. Steps 2 and 3 both use $\beta_{nc} \neq 0$ fixed. Along a sequence $\beta_{nc,n} = n^{-1/2}b$ with $b \neq 0$, $\hat{\beta}_{nc}$ is $O_p(n^{-1/2})$ and no longer converges to a fixed direction, so the continuity of $\beta_{nc} \mapsto P(\beta_{nc})$ invoked in Step 3 is lost at the boundary $\beta_{nc} = 0$: the rank of $\hat{J}^*_{\theta\theta.1}$ remains $k-2$ in every finite sample while the rank of its limit is $k-1$, and the χ^2_{k-2} approximation is not justified. We do not know the correct limiting distribution in this case, and leave it open. Proposition 5 does not close this gap: it is a finite-sample statement, exact under normality at every fixed n regardless of where β_{nc} sits, and it neither uses nor repairs the constant-rank condition just shown to fail.

D Numerical robustness of the exact cancellation

Proposition 3 sets $\tilde{S}_\rho$ and $\tilde{S}_\lambda$ to zero in every sample as an algebraic identity, verified numerically across random draws to within 10^{-14} (Section 4). That check is run at generic, well-conditioned designs, and by construction it says nothing about the corner the proof of Proposition 3 passes through, where the plug-in nuisance block is inverted rather than cancelled by hand. The determinant of the block partialled out for $\tilde{S}_\rho$ and for $\tilde{S}_\lambda$ is $\det(\hat{J}_{\theta\theta.1})\,T_W$, so the quantity that governs any loss of precision there is the condition number of $\hat{J}_{\theta\theta.1}$ together with T_W, not $\hat{J}_{\rho\rho.1} = \hat{\beta}'_{nc}\hat{J}_{\theta\theta.1}\hat{\beta}_{nc} + T_W$; the latter is the (ρ,ρ) entry of the block and governs instead the 2×2 (ρ,λ) inversion behind the surviving $\tilde{S}_\theta$, whose determinant is $(\hat{\beta}'_{nc}\hat{J}_{\theta\theta.1}\hat{\beta}_{nc})\,T_W$.

Table 8 stresses the cancellation along that axis. Near-collinear columns of X_{nc} raise $\mathrm{cond}(\hat{J}_{\theta\theta.1})$ from order one to 10^{12}, and each adjusted score is formed twice: once through the partitioned inverse of the plug-in block, the route the proof formalizes, and once through the identity $\tilde{S}_\rho = S_\rho - S_\lambda - \hat{\beta}'_{nc}S_\theta$. The identity route stays at machine level throughout. The partitioned-inverse route degrades with the conditioning of $\hat{J}_{\theta\theta.1}$, rising from machine level to order 10^{-9} as that condition number reaches 10^{12}, well inside the $\mathrm{cond}(\hat{J}_{\theta\theta.1})$-times-machine-epsilon bound for inverting the block. That level of conditioning is already excluded by the rank condition of Assumption 3, whose recommended practice of monitoring $\mathrm{cond}(X'_{nc}W'MWX_{nc})$ flags it directly. The exactness asserted in Proposition 3 is therefore not a floating-point artifact of well-behaved draws, and the statistics of Theorem 1 are in any case computed from the closed forms, which never invert the block.

Table 8: Numerical departure of the jointly adjusted scores from their exact zero, by conditioning of $\hat{J}_{\theta\theta.1}$. Entries are $\max(|\tilde{S}_\rho|, |\tilde{S}_\lambda|)$ relative to $|S_\rho| + |S_\lambda| + \|\hat{\beta}_{nc}\|\,\|S_\theta\|$, maximum over 200 replications per row, $n = 100$, $k = 3$, row-standardized 6-nearest-neighbour W. The "partitioned inverse" column forms and inverts the plug-in nuisance block of the proof of Proposition 3; the "identity" column uses $\tilde{S}_\rho = S_\rho - S_\lambda - \hat{\beta}'_{nc}S_\theta$. Conditioning is raised by making the columns of X_{nc} near-collinear.

$\mathrm{cond}(\hat{J}_{\theta\theta.1})$	partitioned inverse	identity
3.4×10^{0}	1.2×10^{-14}	1.2×10^{-14}
2.0×10^{2}	3.9×10^{-14}	2.9×10^{-14}
2.2×10^{4}	2.2×10^{-13}	1.2×10^{-14}
2.0×10^{6}	4.5×10^{-12}	1.9×10^{-14}
2.0×10^{8}	3.6×10^{-11}	2.0×10^{-14}
2.0×10^{10}	1.3×10^{-9}	1.5×10^{-13}
2.1×10^{12}	3.9×10^{-9}	5.3×10^{-14}

E Size and power of OS_θ when $E[X'WX_{nc}] \neq 0$

The Monte Carlo design of Section 5.1 imposes $E[X'WX_{nc}] = 0$, which the applications violate. To measure what that costs, we generate the two regressors as spatial autoregressive processes $X_j = (I - \phi W)^{-1}\xi_j$, $\xi_j \sim \mathcal{N}(0, I)$, on the canonical hexagonal lattice at $n = 100$, so that ϕ moves $\widehat{\text{corr}}(X, WX)$ from 0.11 at $\phi = 0$ to 0.85 at $\phi = 0.9$; the remaining settings are those of Section 5.1, with $\beta_{nc} = (1, 2)'$ and 5000 replications, so the Monte Carlo standard error of a 5% rejection rate is 0.003. Table 9 reports, at nominal 5%, the size of OS_θ and of the exact OF_θ at the global null $\theta = 0$; the size of OS_θ under its own null $\theta = \tau\beta_{nc}$, on the naive OLS direction and on the augmented-fit direction; and the power against an off-line alternative $\theta = 0.25(-\beta_{nc,2}, \beta_{nc,1})'$.

Table 9: OS_θ under $E[X'WX_{nc}] \neq 0$. Hexagonal W, $n = 100$, $k = 3$, 5000 replications, nominal 5% (MC s.e. = 0.003). The aligned columns give the size of OS_θ under $\theta = \tau\beta_{nc}$; power is against the off-line θ.

		size ($\theta = 0$)		$\theta = 0.1\beta_{nc}$		$\theta = 0.3\beta_{nc}$		power	
ϕ	$\widehat{\text{corr}}$	OS_θ	OF_θ	naive	aug	naive	aug	naive	aug
0.0	0.11	0.049	0.046	0.052	0.052	0.049	0.048	0.661	0.662
0.3	0.24	0.054	0.052	0.053	0.056	0.044	0.049	0.746	0.746
0.6	0.53	0.054	0.052	0.050	0.058	0.041	0.060	0.854	0.854
0.9	0.85	0.053	0.050	0.054	0.069	0.034	0.066	0.964	0.967

Some readings follow. The size at the global null is flat in the correlation, between 0.046 and 0.054 for both OS_θ and OF_θ across the range, so $E[X'WX_{nc}] \neq 0$ does not distort the null distribution; this is the exactness argument of Section 4.1 at work, since under $\theta = 0$ the OLS fit is unbiased regardless of the correlation. The power against genuine off-line content is likewise unaffected, and if anything rises with ϕ as the regressors carry more spatial variance, with the naive and augmented-fit directions indistinguishable. The only residual is under a strongly aligned effect: at $\phi = 0.9$ the naive OLS direction turns mildly conservative (0.034 at $\tau = 0.3$) while the augmented-fit direction turns mildly liberal (0.066), a few points off nominal either way. The same asymmetry appears in the point estimate, where the mean OS_θ under $\theta = 0.1\beta_{nc}$ holds at its null value of about 1.0 for the naive direction across all ϕ but drifts to 1.16 for the augmented-fit direction at $\phi = 0.9$. The condition $E[X'WX_{nc}] = 0$ therefore bounds a second-order effect on the estimated direction, not the size or the power of the test, and the naive OLS direction, which is the default, is the conservative one.